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Question Bank with Textbook Answers: CIE and SEE papers, 2019-2026

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**Compiled from past papers**	**Maximum Marks**	**–**
Course Code	**IM352IA**	**Duration**	**–**
Sem	**V**	**CIE Question Bank**	
OPERATIONS MANAGEMENT			

Note:

1. Source paper and unit are shown under every question. Part A: fill-in-the-blanks and 2-mark questions. Part B: 6 to 20 mark descriptive questions.
2. **Answers** are quoted word for word from the textbook supplied in the Unit I and Unit II notes (Krajewski, *Operations Management: Processes and Supply Chains*); the page of the book is given in square brackets after each quotation. Where several passages are joined, the omitted words are marked by …. Every quotation was checked automatically against the textbook text.
3. Figures and tables marked "reproduced from the textbook" are cut directly from the book pages. Nothing is redrawn except where a note says so (Part B Q23 and Q25, for which the book has no figure).
4. Grey boxed text **Working** is arithmetic done with the textbook's own formula where the book does not print the solution. Red text **Not in the supplied textbook** marks a point for which the supplied Unit I and Unit II pages contain no answer; no answer from outside the book has been added. A dagger (†) marks a fill-in-the-blank answer where the book has the closest, but not the exact, wording.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Operations ensure the efficient use of _____ to meet customer demands. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Answer: resources

"An operation is a group of resources performing all or part of one or more processes." [p. 22]

Note: † Closest textbook statement; the book does not use the exact wording of the question.

2	A supply chain includes the flow of materials, information, and _____ across organizations. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Answer: services

"Supply chain management is the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand." [p. 22]

3	Effective supply chain management ensures timely delivery and reduces _____. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Answer: cost (waste and unnecessary cost) †

"Each activity in a process should add value to the preceding activities; waste and unnecessary cost should be eliminated." [p. 26]

Note: † Closest textbook statement; the book does not use the exact wording of the question.

4	Competitive priorities like cost, quality, and _____ determine how a company competes in the market. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Answer: time (the fourth group is flexibility)

"We focus on nine broad competitive priorities that fall into the four capability groups of cost, quality, time, and flexibility." [p. 31]

5	The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Not in the supplied textbook: Not covered in the supplied textbook (it has no Industry 4.0 / Fourth Industrial Revolution content), so no textbook answer can be quoted. Part A Q11 of this bank is the companion question.

6	Processes can be categorized as _____ or supporting based on their direct impact on the final product. *Unit I | CIE-I, 25/11/2024*	01	1	1

Answer (quoted from the textbook)

Answer: core

"It also focuses attention on the two main types of processes in the supply chain, namely (1) core processes and (2) support processes." [p. 26]

7	Differentiate between order winner and order qualifier. *Unit I | CIE-I, 25/11/2024*	02	2	2

Answer (quoted from the textbook)

Order winner

"An order winner is a criterion that customers use to differentiate the services or products of one firm from those of another." [p. 31]

"Order winners can include price (which is supported by low-cost operations) and other dimensions of quality, time, and flexibility." [p. 31]

Order qualifier

"Minimal level required from a set of criteria for a firm to do business in a particular market segment is called an order qualifier. Fulfilling the order qualifier will not ensure competitive success; it will only position the firm to compete in the market." [p. 33]

"If a minimum threshold level is not met for an order-qualifying dimension (consistent quality, for example) by a firm, then it would get disqualified from even being considered further by its customers." [p. 33]

8	A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs. 100,000, with total variable costs of Rs. 100 per patient. What is the break-even quantity for this service? *Unit I | CIE-I, 25/11/2024*	02	2	2

Answer (quoted from the textbook)

Answer: 1,000 patients

"The break-even quantity is the volume at which total revenues equal total costs." [p. 51]

$$Q = \frac{F}{p - c} = \frac{100{,}000}{200 - 100} = 1{,}000 \text{ patients}$$

(formula and this exact worked example: Example A.1, Krajewski p. 52)

9	When Amazon decides where to locate its fulfillment centers, it is making a _____ level decision in operations management. *Unit I | CIE-I, 06/11/2025*	01	2	2

Answer (quoted from the textbook)

Answer: strategic

"We follow that with understanding how the design of supply chains and major strategic decisions, such as outsourcing, inventory placement, and locating facilities affect performance." [p. 42]

"Locating abroad is a key decision in the design of supply chains because it affects the flow of materials, information, and employees in support of the firm's core processes." [p. 30]

10	Decision-making models in operations include deterministic, probabilistic, and _____ models. *Unit I | CIE-I, 06/11/2025*	01	1	3

Answer (quoted from the textbook)

Answer: uncertainty † (the textbook's three situations are certainty, uncertainty and risk)

"Using this process, we examine decisions under three different situations: certainty, uncertainty, and risk." [p. 56]

Note: † The textbook does not use the words "deterministic" or "probabilistic". Its three decision-making situations are certainty, uncertainty and risk; the two named in the question correspond to certainty and risk, which leaves uncertainty.

11	The Fourth Industrial Revolution is driven by technologies such as IoT, AI, and _____. *Unit I | CIE-I, 06/11/2025*	01	1	1

Answer (quoted from the textbook)

Not in the supplied textbook: Not covered in the supplied textbook (no Industry 4.0 / Fourth Industrial Revolution content), so no textbook answer can be quoted.

12	When a factory balances machine utilization and customer delivery times, it is managing the _____ responsiveness trade-off. *Unit I | CIE-I, 06/11/2025*	01	2	2

Answer (quoted from the textbook)

Not in the supplied textbook: The wording of this question is not in the supplied textbook pages. The textbook index lists "efficiency and responsiveness" under "Supply chain design" (book pp.~392–395, Chapter 10), but those pages are not part of the units supplied, so the blank cannot be quoted. Index-based answer: efficiency.

13	Walmart's use of RFID and data analytics to track inventory is an example of using _____ as a driver of supply chain performance. *Unit I | CIE-I, 06/11/2025*	01	2	3

Answer (quoted from the textbook)

Answer: information †

"Walmart does more than just distribute Procter & Gamble's products: It shares daily sales information and works with Procter & Gamble in managing inventories and warehousing operations." [p. 121]

"radio frequency identification (RFID) A method for identifying items through the use of radio signals from a tag attached to an item." [Glossary, p. 648]

Note: † The chapter on supply chain drivers (Chapter 10) is not in the supplied units; the answer rests on the Walmart statement above.

14	Zara's ability to rapidly design and distribute new fashions demonstrates the competitive priority of _____. *Unit I | CIE-I, 06/11/2025*	01	2	2

Answer (quoted from the textbook)

Answer: development speed

"Zara is known for its ability to bring fashionable clothing designs from the runway to market quickly." [p. 32]

"Quickly introducing a new service or a product" [p. 32]

15	Strategies for change include process improvement, process redesign, and process _____. *Unit II | CIE-I, 06/11/2025*	01	1	1

Answer (quoted from the textbook)

Answer: reengineering

"We conclude with two different but complementary philosophies for process design: (1) process reengineering and (2) process improvement." [p. 128]

16	Strategic fit refers to the alignment between business strategy and _____ strategy. *Unit II | CIE-I, 06/11/2025*	01	2	2

Answer (quoted from the textbook)

Answer: operations (process) †

"Process strategy guides a variety of process decisions, and in turn is guided by operations strategy and the organization's ability to obtain the resources necessary to support them." [p. 110]

"The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective." [p. 124]

Note: † The book does not give a one-line definition in these words; the answer follows the two statements above.

17	In manufacturing, the four process structures are job process, batch process, line process, and _____ process. *Unit II | CIE-I, 06/11/2025*	01	1	1

Answer (quoted from the textbook)

Answer: continuous-flow

"The manager has four process choices, which form a continuum, to choose from: (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process." [p. 115]

18	In a make-to-order process, production begins only after receiving a _____ from the customer. *Unit II | CIE-I, 06/11/2025*	01	1	1

Answer (quoted from the textbook)

Answer: order (customer order) †

"Typically, they make products to order and do not produce them ahead of time. Each new order is handled as a single unit—as a job." [p. 115]

"Manufacturers that make products to customer specifications in low volumes tend to use the make-to-order strategy" [p. 116]

Note: † Closest textbook statement; the book does not use the exact wording of the question.

19	In services, the process structure is primarily influenced by the level of _____ and customization. *Unit II | CIE-II, 07/01/2025*	01	1	2

Answer (quoted from the textbook)

Answer: customer contact

"A good process strategy for a service process depends first and foremost on the type and amount of customer contact." [pp. 112–113]

"The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics." [p. 113]

20	In manufacturing, a _____ process is ideal for high-volume, standardized production. *Unit II | CIE-II, 07/01/2025*	01	2	2

Answer (quoted from the textbook)

Answer: line (at the extreme, continuous-flow)

"A line process lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products." [p. 116]

"A continuous flow process is the extreme end of high-volume standardized production, with rigid line flows." [p. 116]

21	The first step in a systematic approach to long-term capacity decisions is _____ future demand. *Unit II | CIE-II, 07/01/2025*	01	2	2

Answer (quoted from the textbook)

Answer: estimating (Step 1: estimate future capacity requirements)

"1. Estimate future capacity requirements. 2. Identify gaps by comparing requirements with available capacity. 3. Develop alternative plans for reducing the gaps. 4. Evaluate each alternative, both qualitatively and quantitatively, and make a final choice." [p. 228]

"Either way, the foundation for the estimate is forecasts of demand, productivity, competition, and technological change." [p. 228]

22	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____. *Unit II | CIE-II, 07/01/2025*	01	1	2

Answer (quoted from the textbook)

Answer: alternatives

"4. Evaluate each alternative, both qualitatively and quantitatively, and make a final choice." [p. 228]

23	_____ ensures that operational capabilities are in line with the company's strategic goals. *Unit II | CIE-II, 07/01/2025*	01	2	3

Answer (quoted from the textbook)

Answer: operations strategy †

"Operations strategy specifies the means by which operations implements corporate strategy and helps to build a customer-driven firm. It links long-term and short-term operations decisions to corporate strategy and develops the capabilities the firm needs to be competitive." [p. 27]

Note: † Closest textbook statement; the book does not use the exact wording of the question.

24	A _____ shop deals with customized, low-volume production. *Unit II | Improvement CIE, 12/01/2026*	01	1	2

Answer (quoted from the textbook)

Answer: job

"A job process creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low." [p. 115]

25	In mass production, the layout used is _____ layout. *Unit II | Improvement CIE, 12/01/2026*	01	1	2

Answer (quoted from the textbook)

Answer: line (product-oriented line flow) †

"Combining a line process with the make-to-stock strategy is sometimes called mass production." [p. 117]

"Organizing around the product means bringing together all the different human resources and equipment needed for a specific product and dedicating them to producing just that product." [p. 115]

Note: † The book does not use the term "product layout"; it describes mass production as a line process.

26	The maximum output a system can produce is called its _____. *Unit II | Improvement CIE, 12/01/2026*	01	1	2

Answer (quoted from the textbook)

Answer: capacity

"Capacity is the maximum rate of output of a process or a system." [p. 222]

27	The long-term decision of choosing a geographic site for operations is called _____. *Unit II | Improvement CIE, 12/01/2026*	01	2	3

Answer (quoted from the textbook)

Answer: facility location (locating facilities) †

"We follow that with understanding how the design of supply chains and major strategic decisions, such as outsourcing, inventory placement, and locating facilities affect performance." [p. 42]

Note: † The location chapter (Chapter 11) is not in the supplied units; the answer rests on the statement above.

28	Continuous processes are suitable for _____ production. *Unit II | Improvement CIE, 12/01/2026*	01	2	3

Answer (quoted from the textbook)

Answer: high-volume, standardized production

"A continuous flow process is the extreme end of high-volume standardized production, with rigid line flows." [p. 116]

"Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food" [p. 116]

Sl. No	Questions	M	BT	CO
	Part – B			
1	Define operations management. Briefly explain the key terms involved in the definition of operations management with a schematic model of a production system. *Unit I | CIE-I, 06/11/2025 and 25/11/2024 (same question)*	10	2	1

Answer (quoted from the textbook)

Definition

"Operations management refers to the systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external customers." [p. 22]

Key terms in the definition

"A process is any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers. For organizational purposes, processes tend to be clustered together into operations. An operation is a group of resources performing all or part of one or more processes. Processes can be linked together to form a supply chain, which is the interrelated series of processes within a firm and across different firms that produce a service or product to the satisfaction of customers." [p. 22]

"Supply chain management is the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand." [p. 22]

"Any process has inputs and outputs. Inputs can include a combination of human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land, and energy." [p. 24]

"Processes provide outputs to customers. These outputs may often be services (that can take the form of information) or tangible products. Every process and every person in an organization has customers. Some are external customers, who may be end users or intermediaries (e.g., manufacturers, financial institutions, or retailers) buying the firm's finished services or products. Others are internal customers, who may be employees in the firm whose process inputs are actually the outputs of earlier processes managed within the firm." [p. 24]

"In a similar fashion, every process and every person in an organization relies on suppliers. External suppliers may be other businesses or individuals who provide the resources, services, products, and materials for the firm's short-term and long-term needs. Processes also have internal suppliers, who may be employees or processes that supply important information or materials." [p. 24]

Schematic model

▼ **FIGURE 1.2**
Processes and Operations

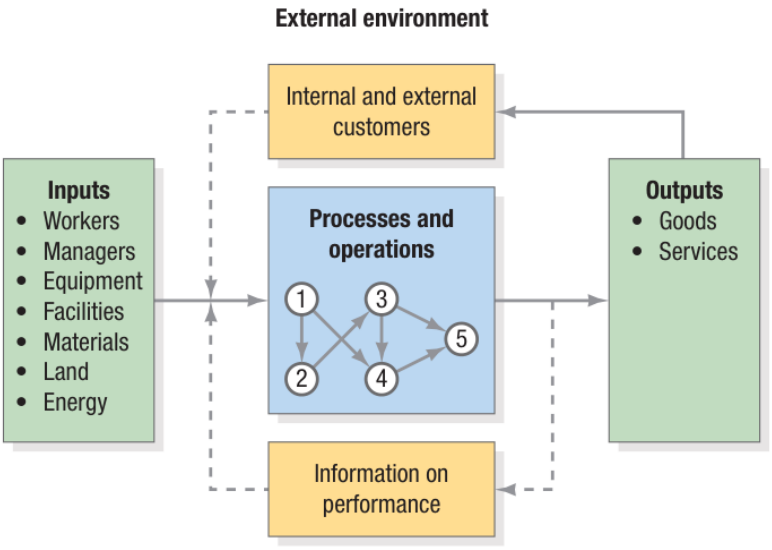


Figure 1.2: Processes and Operations (reproduced from the textbook)

"The numbered circles in Figure 1.2 represent operations through which services, products, or customers pass and where processes are performed. The arrows represent flows, and can cross because one job or customer can have different requirements (and thus a different flow pattern) than the next job or customer." [p. 24]

"The dashed lines represent two special types of input: participation by customers and information on performance from both internal and external sources." [p. 25]

2	With suitable example, distinguish between manufacturing and service processes. *Unit I | CIE-I, 06/11/2025 and 25/11/2024 (same question)*	10	2	2

Answer (quoted from the textbook)

"While Figure 1.3 shows several distinctions between service and manufacturing processes along a continuum, the two key differences that we discuss in detail are (1) the nature of their output and (2) the degree of customer contact." [p. 25]

Manufacturing process

"Manufacturing processes convert materials into goods that have a physical form we call products. For example, an assembly line produces a 350 Z sports car, and a tailor produces an outfit for the rack of an upscale clothing store. The transformation processes change the materials on one or more of the following dimensions:" [p. 25]

(1) "Physical properties" [p. 25]

(2) "Shape"

(3) "Size (e.g., length, breadth, and height of a rectangular block of wood)" [p. 25]

(4) "Surface finish" [p. 25]

(5) "Joining parts and materials" [p. 25]

"The outputs from manufacturing processes can be produced, stored, and transported in anticipation of future demand." [p. 25]

"Manufacturing processes tend to have less customer contact. For example, washing machines are ultimately produced to meet retail forecasts." [p. 26]

Service process

"If a process does not change the properties of materials on at least one of these five dimensions, it is considered a service (or nonmanufacturing) process. Service processes tend to produce intangible, perishable outputs. For example, the output from the auto loan process of a bank would be a car loan, and an output of the order fulfillment process of the U.S. Postal Service is the delivery of your letter. The outputs of service processes typically cannot be held in a finished goods inventory to insulate the process from erratic customer demands." [p. 25]

"Service processes tend to have a higher degree of customer contact. Customers may take an active role in the process itself, as in the case of shopping in a supermarket, or they may be in close contact with the service provider to communicate specific needs, as in the case of a medical clinic." [p. 26]

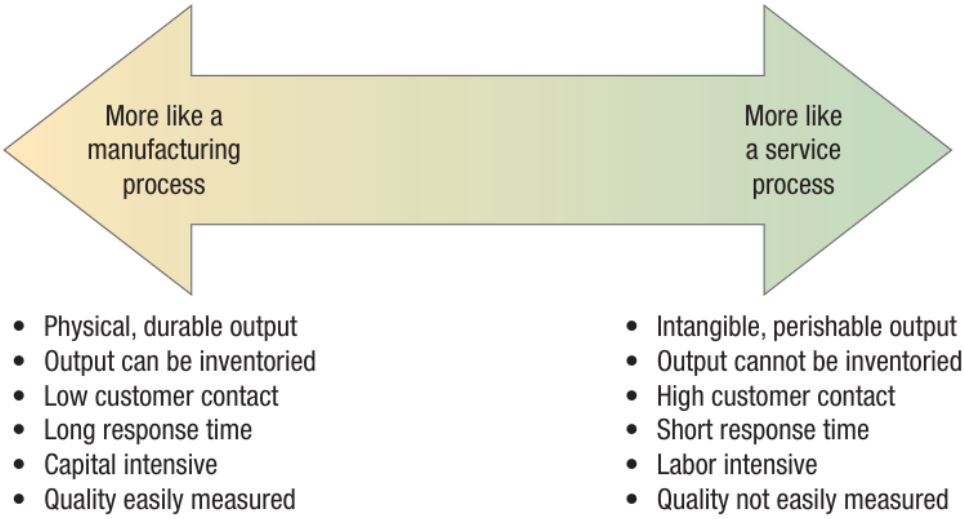


Figure 1.3: Continuum of Characteristics of Manufacturing and Service Processes (reproduced from the textbook)

"In general, manufacturing processes also have longer response times, are more capital intensive, and their quality can be measured more easily than those of service processes." [p. 25]

Example from the textbook

"For example, the process of cooking a hamburger at a McDonald's is a manufacturing process because it changes the material's physical properties (dimension 1), as is the process of assembling the hamburger with the bun (dimension 5). However, most of the other processes visible or invisible to McDonald's customers are service processes." [p. 26]

3	A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff? *Unit I | CIE-I, 06/11/2025*	16	4	3

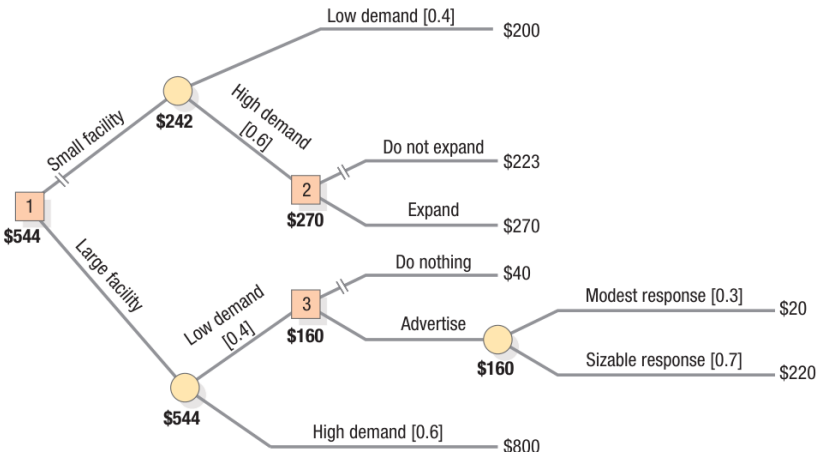
Answer (quoted from the textbook)

Decision tree: nodes and branches

"A decision tree is a schematic model of alternatives available to the decision maker, along with their possible consequences. The name derives from the tree-like appearance of the model. It consists of a number of square nodes, representing decision points, which are left by branches (which should be read from left to right), representing the alternatives. Branches leaving circular, or chance, nodes represent the events. The probability of each chance event, P(E), is shown above each branch. The probabilities for all branches leaving a chance node must sum to 1.0." [p. 59]

Decision tree for the retailer (Figure A.5, textbook)

"The decision tree in Figure A.5 shows the event probability and the payoff for each of the seven alternative-event combinations. The first decision is whether to build a small or a large facility. Its node is shown first, to the left, because it is the decision the retailer must make now. The second decision node—whether to expand at a later date—is reached only if a small facility is built and demand turns out to be high. Finally, the third decision point—whether to advertise—is reached only if the retailer builds a large facility and demand turns out to be low." [p. 60]



◀ **FIGURE A.5**
Decision Tree for Retailer (in $000)

Figure A.5: Decision Tree for Retailer (in $000) (reproduced from the textbook)

Analysis (rolling back from right to left)

"After drawing a decision tree, we solve it by working from right to left, calculating the expected payoff for each node as follows: 1. For an event node, we multiply the payoff of each event branch by the event's probability. We add these products to get the event node's expected payoff. 2. For a decision node, we pick the alternative that has the best expected payoff." [p. 60]

(1) "For the event node dealing with advertising, the expected payoff is 160, or the sum of each event's payoff weighted by its probability [0.3(20) + 0.7(220)]." [p. 60]

(2) "The expected payoff for decision node 3 is 160 because Advertise (160) is better than Do nothing (40). Prune the Do nothing alternative." [p. 60]

(3) "The payoff for decision node 2 is 270 because Expand (270) is better than Do not expand (223). Prune Do not expand." [p. 60]

(4) "The expected payoff for the event node dealing with demand, assuming that a small facility is built, is 242 [or 0.4(200) + 0.6(270)]." [p. 60]

(5) "The expected payoff for the event node dealing with demand, assuming that a large facility is built, is 544 [or 0.4(160) + 0.6(800)]." [p. 60]

(6) "The expected payoff for decision node 1 is 544 because the large facility's expected payoff is largest. Prune Small facility." [p. 60]

Decision

"The retailer should build the large facility. This initial decision is the only one made now. Subsequent decisions are made after learning whether demand actually is low or high." [p. 61]

Answer: Large facility: expected payoff $544,000 (small facility: $242,000).

4	A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the payoff table below. What is the best alternative for each decision rule: Maximin, Maximax, Laplace, Minimax Regret?	14	4	3

Alternative	Low demand	High demand
Small facility	200	270
Large facility	160	800
Do nothing	0	0

Unit I | CIE-I, 06/11/2025 (14 M); CIE-I, 25/11/2024 (10 M)

Answer (quoted from the textbook)

"The "do nothing" alternative is dominated by the other alternatives; that is, the outcome of one alternative is no better than the outcome of another alternative for each event. Because the "do nothing" alternative is dominated, the manager does not consider it further." [p. 57]

Decision rules (textbook definitions)

"Maximin. Choose the alternative that is the "best of the worst." This rule is for the pessimist, who anticipates the "worst case" for each alternative." [p. 57]

"Maximax. Choose the alternative that is the "best of the best." This rule is for the optimist who has high expectations and prefers to "go for broke."" [p. 57]

"Laplace. Choose the alternative with the best weighted payoff. To find the weighted payoff, give equal importance (or, alternatively, equal probability) to each event. If there are n events, the importance (or probability) of each is 1/n, so they add up to 1.0. This rule is for the realist." [p. 57]

"Minimax Regret. Choose the alternative with the best "worst regret." Calculate a table of regrets (or opportunity losses), in which the rows represent the alternatives and the columns represent the events. A regret is the difference between a given payoff and the best payoff in the same column." [p. 57]

Solution (Example A.6, payoffs in $000)

"An alternative's worst payoff is the lowest number in its row of the payoff matrix, because the payoffs are profits." [p. 57]

Alternative	Worst payoff (Maximin)	Best payoff (Maximax)
Small facility	200	270
Large facility	160	800

"The best of these worst numbers is $200,000, so the pessimist would build a small facility." [p. 57]

"The best of these best numbers is $800,000, so the optimist would build a large facility." [p. 58]

"With two events, we assign each a probability of 0.5. Thus, the weighted payoffs ($000) are" [p. 58]

Alternative	Weighted payoff (Laplace)
Small facility	0.5(200) + 0.5(270) = 235
Large facility	0.5(160) + 0.5(800) = 480

"The best of these weighted payoffs is $480,000, so the realist would build a large facility." [p. 58]

"If demand turns out to be low, the best alternative is a small facility and its regret is 0 (or 200 − 200). If a large facility is built when demand turns out to be low, the regret is 40 (or 200 − 160)." [p. 58]

Alternative	Low demand	High demand	Maximum regret
Small facility	200 − 200 = 0	800 − 270 = 530	530
Large facility	200 − 160 = 40	800 − 800 = 0	40

Regret table (reproduced from Example A.6)

"The column on the right shows the worst regret for each alternative. To minimize the maximum regret, pick a large facility. The biggest regret is associated with having only a small facility and high demand." [p. 58]

Decision

"The pessimist would choose the small facility. The realist, optimist, and manager choosing to minimize the maximum regret would build the large facility." [p. 58]

Decision rule	Best alternative
Maximin	Small facility
Maximax	Large facility
Laplace	Large facility
Minimax regret	Large facility

5	White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is \$50,000. The annual cost of installing two lifts is only \$90,000 if both are purchased at the same time. If used at all, each lift costs \$200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost \$20 per customer per day. Should the resort purchase one lift or two? *Unit I | CIE-I, 25/11/2024*	20	4	3

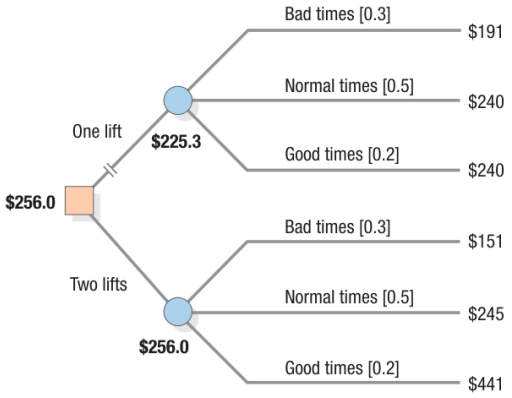
Answer (quoted from the textbook)

Solution (Solved Problem 3, textbook)

"The decision tree is shown in Figure A.6. The payoff (\$000) for each alternative-event branch is shown in the following table. The total revenues from one lift operating at 100 percent capacity are \$490,000 (or 250 customers × 98 days × \$20/customer-day)." [p. 64]

Alternative	Economic Condition	Payoff Calculation (Revenue – Cost)
One lift	Bad times	0.9(490) – (50 + 200) = 191
	Normal times	1.0(490) – (50 + 200) = 240
	Good times	1.0(490) – (50 + 200) = 240
Two lifts	Bad times	0.9(490) – (90 + 200) = 151
	Normal times	1.5(490) – (90 + 400) = 245
	Good times	1.9(490) – (90 + 400) = 441

FIGURE A.6 ▶



Payoff table and Figure A.6: decision tree for the ski lifts (reproduced from the textbook, in \$000)

Rolling back the tree (textbook rule)

"For an event node, we multiply the payoff of each event branch by the event's probability. We add these products to get the event node's expected payoff." [p. 60]

$$\text{One lift: } 0.3(191) + 0.5(240) + 0.2(240) = \$225.3 \text{ (in \$000)}$$

$$\text{Two lifts: } 0.3(151) + 0.5(245) + 0.2(441) = \$256.0 \text{ (in \$000)}$$

Working (textbook method applied to the numbers of the question)

"For a decision node, we pick the alternative that has the best expected payoff." [p. 60]

Figure A.6 shows \$225.3 on the one-lift event node and \$256.0 on the two-lift event node; the branch with the lower value (one lift) is marked as pruned.

Answer: Purchase two lifts: expected payoff \$256,000 against \$225,300 for one lift.

6	Explain the major process strategy decisions and their implications for operations. *Unit II | CIE-II, 07/01/2025*	10	2	2

Answer (quoted from the textbook)

"We begin by defining four basic process decisions: (1) process structure (including layout), (2) customer involvement, (3) resource flexibility, and (4) capital intensity. We discuss these decisions for both service and manufacturing processes." [p. 110]

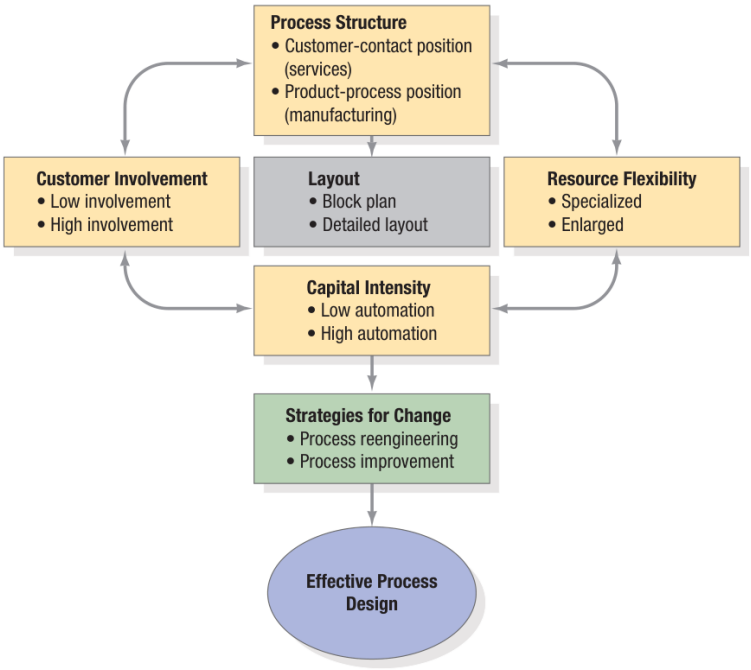


Figure 3.1: Major Decisions for Effective Processes (reproduced from the textbook)

The four decisions

"Process structure determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A layout, which is the physical arrangement of operations created from the various processes, puts these decisions into tangible form." [p. 112]

"Customer involvement reflects the ways in which customers become part of the process and the extent of their participation." [p. 112]

"Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions." [p. 112]

"Capital intensity is the mix of equipment and human skills in a process. The greater the relative cost of equipment, the greater is the capital intensity." [p. 112]

Implications for operations

"The key to successful process decisions is to make choices that fit the situation and that make sense together. They should not work at cross-purposes, with one process optimized at the expense of other processes." [p. 110]

"High customer contact at a front-office service process means: 1. Process Structure. The customer (internal or external) is present, actively involved, and receives personal attention. These conditions create processes with high divergence and flexible process flows. 2. Customer Involvement. When customer contact is high, customers are more likely to become part of the process. The service created for each customer is unique. 3. Resource Flexibility. High process divergence and flexible process flows fit with more flexibility from the process's resources—its workforce, facilities, and equipment. 4. Capital Intensity. When volume is higher, automation and capital intensity are more likely." [p. 125]

"High volumes per part type at a manufacturing process typically mean: 1. Process Structure. High volumes, combined with a standard product, make a line flow possible. It is just the opposite where a job process produces to specific customer orders. 2. Customer Involvement. Customer involvement is not a factor in most manufacturing processes, except for choices made on product variety and customization." [p. 126]

"Resource Flexibility. When volumes are high and process divergence is low, flexibility is not needed to utilize resources effectively, and specialization can lead to more efficient processes. 4. Capital Intensity. High volumes justify the large fixed costs of an efficient operation." [p. 127]

7	Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices.	10	3	2

Department	A	B	C	D	E	F
A	–	25	90			185
B		–			105	
C			–		125	125
D				–	25	
E					–	105
F						–

a. If other factors are equal, which two offices should be located closest together?
b. A figure shows an alternative layout for the department: top row C, F, A; bottom row B, E, D. What is the total weighted-distance score for this plan based on rectilinear distance, assuming that offices A and B are three units of distance apart?

Unit II | CIE-II, 07/01/2025

Answer (quoted from the textbook)

(a) Which two offices should be closest together?

"For example, the most important interaction is between the administration and internal audit departments for OBM, with a score of 10. This closeness factor is given in the first row and last column. Thus, the designer should locate departments 1 and 6 close together" [p. 118]

In Garber's matrix the largest value is 185 trips, between offices A and F.

Answer: Offices A and F should be located closest together (185 trips).

(b) Weighted-distance score of the layout in Figure 3.16

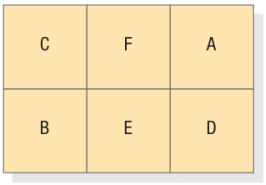
"The layout designer seeks to minimize the weighted-distance (wd) score by locating centers that have high-closeness ratings close together. To calculate a layout's wd score, we use either of the distance measures and simply multiply

the proximity scores by the distances between centers. The sum of those products becomes the layout's final wd score—the lower the better." [p. 119]

"Rectilinear distance measures the distance between two points with a series of 90-degree turns, as along city blocks. The distance traveled in the x-direction is the absolute value of the difference between the x-coordinates. Adding this result to the absolute value of the difference between the y-coordinates gives" [p. 119]

$$d_{AB} = |x_A - x_B| + |y_A - y_B|$$

(rectilinear distance, Krajewski p. 119)



◀ **FIGURE 3.16**
Alternative Block Plan

Figure 3.16: Alternative Block Plan (reproduced from the textbook)

Adjacent offices are one unit apart, so A and B (top-right and bottom-left) are 2 + 1 = 3 units apart, as the question states.

Pair	Trips (w)	Rectilinear distance (d)	wd = w × d
A, B	25	3	75
A, C	90	2	180
A, F	185	1	185
B, E	105	1	105
C, E	125	2	250
C, F	125	1	125
D, E	25	1	25
E, F	105	1	105
		Total	1,050

Answer: Total weighted-distance score of the plan = 1,050.

Note: The textbook gives the method (quoted above) but not the solution of this problem; the table applies that method to the matrix and plan in the question.

8	Explain the concepts of economies and diseconomies of scale with appropriate examples. *Unit II | Improvement CIE, 12/01/2026*	10	2	2

Answer (quoted from the textbook)

Economies of scale

"A concept known as economies of scale states that the average unit cost of a service or good can be reduced by increasing its output rate. Four principal reasons explain why economies of scale can drive costs down when output increases: (1) Fixed costs are spread over more units; (2) construction costs are reduced; (3) costs of purchased materials are cut; and (4) process advantages are found." [p. 224]

(1) "In the short term, certain costs do not vary with changes in the output rate. These fixed costs include heating costs, debt service, and managers' salaries. The depreciation of plant and equipment already owned is also a fixed cost in the accounting sense. When the average output rate—and, therefore, the facility's utilization rate—increases, the average unit cost drops because fixed costs are spread over more units." [p. 224]

(2) "Certain activities and expenses are required to build small and large facilities alike: building permits, architects' fees, and rental of building equipment. Doubling the size of the facility usually does not double construction costs." [p. 224]

(3) "Higher volumes can reduce the costs of purchased materials and services. They give the purchaser a better bargaining position and the opportunity to take advantage of quantity discounts. Retailers such as Walmart reap significant economies of scale because their national and international stores buy and sell huge volumes of each item." [p. 224]

(4) "High-volume production provides many opportunities for cost reduction. At a higher output rate, the process shifts toward a line process, with resources dedicated to individual products. Firms may be able to justify the expense of more efficient technology or more specialized equipment. The benefits from dedicating resources to individual services or products may include speeding up the learning effect, lowering inventory, improving process and job designs, and reducing the number of changeovers." [p. 224]

Diseconomies of scale

"Bigger is not always better, however. At some point, a facility can become so large that diseconomies of scale set in; that is, the average cost per unit increases as the facility's size increases. The reason is that excessive size can bring complexity, loss of focus, and inefficiencies that raise the average unit cost of a service or product." [p. 224]

"Too many layers of employees and bureaucracy can cause management to lose touch with employees and customers. A less agile organization loses the flexibility needed to respond to changing demand." [p. 224]

Example: hospitals (Figure 6.1)

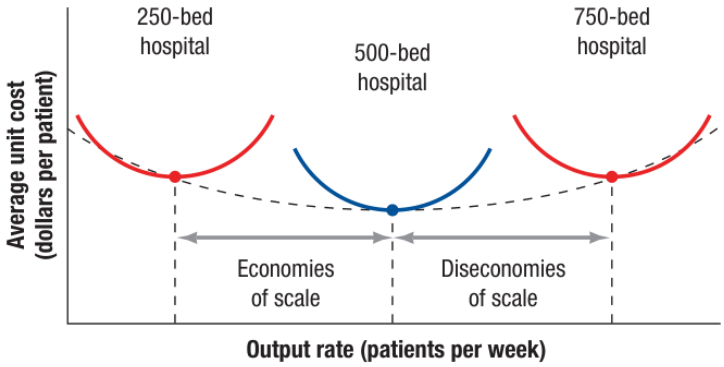


Figure 6.1: Economies and Diseconomies of Scale (reproduced from the textbook)

"Figure 6.1 illustrates the transition from economies of scale to diseconomies of scale. The 500-bed hospital shows economies of scale because the average unit cost at its best operating level, represented by the blue dot, is less than that of the 250-bed hospital. However, assuming that sufficient demand exists, further expansion to a 750-bed hospital leads to higher average unit costs and diseconomies of scale." [p. 225]

"Figure 6.1 does not mean that the optimal size for all hospitals is 500 beds. Optimal size depends on the number of patients per week to be served. On the one hand, a hospital serving a small community could have lower costs by choosing a 250-bed capacity rather than the 500-bed capacity. On the other hand, a large community might be served more efficiently by two 500-bed hospitals than by one 1,000-bed facility if diseconomies of scale exist at the bigger size." [p. 225]

9	Explain process reengineering and process improvement, highlighting their differences with suitable examples. *Unit II | Improvement CIE, 12/01/2026*	10	4	3

Answer (quoted from the textbook)

"We conclude with two different but complementary philosophies for process design: (1) process reengineering and (2) process improvement." [p. 128]

Process reengineering

"Reengineering is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed. Process reengineering is about reinvention, rather than incremental improvement. It is strong medicine and not always needed or successful. Pain, in the form of layoffs and large cash outflows for investments in information technology, almost always accompanies massive change. However, reengineering processes can have big payoffs." [p. 128]

Process improvement

"Process improvement is the systematic study of the activities and flows of each process to improve it. Its purpose is to "learn the numbers," understand the process, and dig out the details. Once a process is really understood, it can be improved." [p. 129]

"One must look for ways to streamline tasks, eliminate whole processes entirely, cut expensive materials or services, improve the environment, or make jobs safer. One must find the ways to trim costs and delays and to improve customer satisfaction." [p. 129]

Differences (each cell quoted from the textbook)

Aspect	Process reengineering	Process improvement
Meaning	Reengineering is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed.	Process improvement is the systematic study of the activities and flows of each process to improve it.
Scale of change	Process reengineering is about reinvention, rather than incremental improvement.	Process improvement goes on, whether or not a process is reengineered. There is always a better way.
Cost and risk	Pain, in the form of layoffs and large cash outflows for investments in information technology, almost always accompanies massive change.	One must find the ways to trim costs and delays and to improve customer satisfaction.
Who carries it out	A team, consisting of members from each functional area affected by the process change, is charged with carrying out a reengineering project.	An individual or a whole team examines the process, using the tools described in the next chapter.
Approach	Information technology is a primary enabler of process engineering.	One must look for ways to streamline tasks, eliminate whole processes entirely, cut expensive materials or services, improve the environment, or make jobs safer.

[pp. 128–129]

Examples given in the textbook

"Most reengineering projects design processes around information flows, such as customer order fulfillment." [p. 128]

"The best understanding of a process, and how to improve it, often lies with the people who perform the work each day, not with cross-functional teams or top management." [p. 128]

TABLE 3.4 | KEY ELEMENTS OF REENGINEERING

Element	Description
Critical processes	The emphasis of reengineering should be on core business processes. Normal process-improvement activities can be continued with the other processes.
Strong leadership	Senior executives must provide strong leadership for reengineering to be successful. Otherwise, cynicism, resistance ("we tried that before"), and boundaries between departments can block radical changes.
Cross-functional teams	A team, consisting of members from each functional area affected by the process change, is charged with carrying out a reengineering project. Self-managing teams and employee empowerment are the rule rather than the exception.
Information technology	Information technology is a primary enabler of process engineering. Most reengineering projects design processes around information flows, such as customer order fulfillment.
Clean-slate philosophy	Reengineering requires a "clean-slate" philosophy—that is, starting with the way the customer wants to deal with the company. To ensure a customer orientation, teams begin with internal and external customer objectives for the process.
Process analysis	Despite the clean-slate philosophy, a reengineering team must understand things about the current process: what it does, how well it performs, and what factors affect it. The team must look at every procedure involved in the process throughout the organization.

Table 3.4: Key Elements of Reengineering (reproduced from the textbook)

Note: The textbook gives no separate worked example for process improvement beyond the activities quoted above.

10	Describe a systematic approach to process analysis with the help of a block diagram. *Unit II | Improvement CIE, 12/01/2026*	10	2	2

Answer (quoted from the textbook)

"In Part 2, we focus on analyzing processes and how they can be improved to meet the goals of the operations strategy. We begin by addressing the strategic aspects of process design and then present a six-step systematic approach to process analysis. Each chapter in this part deals with some aspect of that approach. We discuss the tools that help managers analyze processes, and we reveal the methods firms use to measure process performance and quality." [p. 41]

"process analysis The documentation and detailed understanding of how work is performed and how it can be redesigned." [Glossary, p. 647]

"Process improvement is the systematic study of the activities and flows of each process to improve it. Its purpose is to "learn the numbers," understand the process, and dig out the details. Once a process is really understood, it can be improved." [p. 129]

Not in the supplied textbook: The six steps of the systematic approach and its block diagram are in Chapter 4, "Analyzing Processes" (the textbook index lists "Process analysis, 139–160"). That chapter is not in the units supplied, so the steps and the diagram cannot be quoted or reproduced here. Only the statements above are available.

11	A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.	10	3	3

Item	Client X	Client Y
Annual demand forecast (copies)	2,000	6,000
Standard processing time (hours/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time (hours)	0.25	0.40

Unit II | Improvement CIE, 12/01/2026

Answer (quoted from the textbook)

"Setup time is the time required to change a process or an operation from making one service or product to making another." [p. 229]

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1 - (C/100)]}$$

(capacity requirement for multiple products, Krajewski p. 229)

where D = annual demand, p = processing time per unit, Q = lot size, s = setup time per lot, N = hours available per year from one machine, C = desired capacity cushion (%).

Solution (Example 6.1, textbook)

$$M = \frac{[2{,}000(0.5) + (2{,}000/20)(0.25)]_X + [6{,}000(0.7) + (6{,}000/30)(0.40)]_Y}{[(250\text{ day/year})(1\text{ shift/day})(8\text{ hours/day})]\,[1.0 - (15/100)]} = \frac{5{,}305}{1{,}700} = 3.12$$

(Example 6.1, Krajewski p. 230)

Client X: $2{,}000(0.5) + (2{,}000/20)(0.25) = 1{,}000 + 25 = 1{,}025$ h

Client Y: $6{,}000(0.7) + (6{,}000/30)(0.40) = 4{,}200 + 80 = 4{,}280$ h

Total $5{,}305$ h; $\quad N[1 - (C/100)] = 2{,}000(0.85) = 1{,}700$ h per machine

Working (textbook method applied to the numbers of the question)

"What to do when M is not an integer depends on the situation. For example, it is impossible to buy a fractional machine. In this case, round up the fractional part, unless it is cost efficient to use short-term options, such as overtime or stockouts, to cover any shortfalls." [p. 229]

"Rounding up to the next integer gives a requirement of four machines." [p. 230]

"The copy center's capacity is being stretched and no longer has the desired 15 percent capacity cushion with the existing three machines. Not wanting customer service to suffer, management decided to use overtime as a short-term solution to handle past-due orders. If demand continues at the current level or grows, it will acquire a fourth machine." [p. 230]

Answer: Four machines are needed (3.12 rounded up); the centre has three.

12	Consider amazon.com, whose Website enjoys millions of “hits" each day and puts customers in touch with millions of services and products. What are amazon.com's competitive priorities and what should its operations strategy focus on? *Unit I | SEE Jan/Feb 2021 (18IM54), Q2a*	06	–	–

Answer (quoted from the textbook)

Competitive priority of Amazon.com (textbook, Table 1.1)

“Handling a wide assortment of services or products efficiently” [p. 32]

“Amazon.com uses information technology and streamlined customer relationship and order fulfillment processes to reliably deliver a vast variety of items to its customers.” [p. 32]

“Processes supporting variety must be capable of larger volumes than processes supporting customization. Services or products are not necessarily unique to specific customers and may have repetitive demands.” [p. 32]

“Examples include a hotel’s reservation handling, a new car design for an auto manufacturer, or Web-based purchasing for an online retailer like amazon.com.” [p. 27]

What the operations strategy should focus on

“Operations strategy specifies the means by which operations implements corporate strategy and helps to build a customer-driven firm.” [p. 27]

“Any gap between a competitive priority and the capability to achieve that competitive priority must be closed by an effective operations strategy.” [p. 35]

“Developing capabilities and closing gaps is the thrust of operations strategy.” [p. 35]

Note: The textbook poses this as Discussion Question 2 of Chapter 1 and does not give an answer; the statements above are the only Amazon.com-specific material in the supplied chapter.

13	Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities. *Unit I | SEE Feb/Mar 2025 (IM352IA), Q2a*	10	2	1

Answer (quoted from the textbook)

“Competitive priorities are the critical operational dimensions a process or supply chain must possess to satisfy internal or external customers, both now and in the future.” [p. 31]

“We focus on nine broad competitive priorities that fall into the four capability groups of cost, quality, time, and flexibility.” [p. 31]

Competitive priority	Definition	Example
1. Low-cost operations	Delivering a service or a product at the lowest possible cost to the satisfaction of external or internal customers of the process or supply chain	Costco achieves low costs by designing all processes for efficiency, stacking products on pallets in warehouse-type stores, and negotiating aggressively with their suppliers.
2. Top quality	Delivering an outstanding service or product	Rolex is known globally for creating precision timepieces.
3. Consistent quality	Producing services or products that meet design specifications on a consistent basis	McDonald's standardizes work methods, staff training processes, and procurement of raw materials to achieve the same consistent product and process quality from one store to the next.
4. Delivery speed	Quickly filling a customer's order	Dell engineered its customer relationship, order fulfillment, and supplier relationship processes to create an integrated and an agile supply chain that delivers reliable and inexpensive computers to its customers with short lead times.
5. On-time delivery	Meeting delivery-time promises	United Parcel Services (UPS) uses its expertise in logistics and warehousing processes to deliver a very large volume of shipments on-time across the globe.
6. Development speed	Quickly introducing a new service or a product	Zara is known for its ability to bring fashionable clothing designs from the runway to market quickly.
7. Customization	Satisfying the unique needs of each customer by changing service or product designs	Ritz Carlton customizes services to individual guest preferences.
8. Variety	Handling a wide assortment of services or products efficiently	Amazon.com uses information technology and streamlined customer relationship and order fulfillment processes to reliably deliver a vast variety of items to its customers.
9. Volume flexibility	Accelerating or decelerating the rate of production of services or products quickly to handle large fluctuations in demand	The United States Post Office (USPS) can have severe demand peak fluctuations at large postal facilities where processes are flexibly designed for receiving, sorting, and dispatching mail to numerous branch locations.

Table 1.1 (definition and example columns), Krajewski [p. 32]

A consistent pattern of decisions develops capabilities

"Even if it is not formally stated, the current operations strategy for any firm is really the pattern of decisions that have been made for its processes and supply chains." [p. 35]

"Once managers determine the competitive priorities for a process, it is necessary to assess the competitive capabilities of the process. Any gap between a competitive priority and the capability to achieve that competitive priority must be closed by an effective operations strategy." [p. 35]

"Developing capabilities and closing gaps is the thrust of operations strategy." [p. 35]

"The procedure for assessing an operations strategy begins with identifying good measures for each priority. The more quantitative the measures are, the better. Data are gathered for each measure to determine the current capabilities of the process. Gaps are identified by comparing each capability to management's target values for the measures, and unacceptable gaps are closed by appropriate actions." [p. 35]

"Competitive priorities must change and evolve over time along with changing business conditions and customer preferences." [p. 31]

14	Describe the process view of operations in typical organizations with the help of a block diagram. *Unit I | SEE Feb/Mar 2025 (IM352IA), Q2b*	06	2	1

Answer (quoted from the textbook)

"The reason is that a process view of the firm provides a much more relevant picture of the way firms actually work." [p. 24]

"The concept of a process, however, can be much broader. A process can have its own set of objectives, involve a work flow that cuts across departmental boundaries, and require resources from several departments." [p. 24]

"A process is any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers." [p. 22]

▼ **FIGURE 1.2**
Processes and Operations

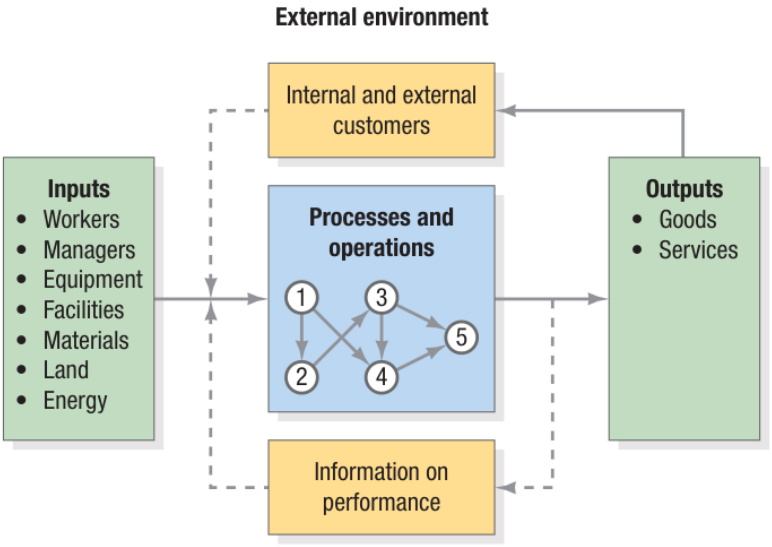


Figure 1.2: Processes and Operations (reproduced from the textbook)

"Any process has inputs and outputs. Inputs can include a combination of human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land, and energy." [p. 24]

"Figure 1.2 can represent a whole firm, a department, a small group, or even a single individual. Each one has inputs and uses processes at various operations to provide outputs." [p. 25]

"You will notice that the key to success in many organizations is a keen understanding of how their processes work, since an organization is only as effective as its processes." [p. 24]

15	Differentiate between competitive priorities and competitive capabilities with suitable examples. *Unit I | SEE Apr/May 2024 (21IM53), Q2a*	08	2	1

Answer (quoted from the textbook)

Competitive priorities

"Competitive priorities are the critical operational dimensions a process or supply chain must possess to satisfy internal or external customers, both now and in the future. Competitive priorities are planned for processes and the supply chain created from them. They must be present to maintain or build market share or to allow other internal processes to be successful. Not all competitive priorities are critical for a given process; management selects those that are most important." [p. 31]

Competitive capabilities

"Competitive capabilities are the cost, quality, time, and flexibility dimensions that a process or supply chain actually possesses and is able to deliver. When the capability falls short of the priority attached to it, management must find ways to close the gap or else revise the priority." [p. 31]

Examples

"Costco achieves low costs by designing all processes for efficiency, stacking products on pallets in warehouse-type stores, and negotiating aggressively with their suppliers." [p. 32]

"Rolex is known globally for creating precision timepieces." [p. 32]

"The credit card division shows significant gaps in the process's capability for low-cost operations. Management's remedy is to redesign the process in ways that reduce costs but will not impair the other competitive priorities." [p. 35]

Competitive priority	Measure	Capability / gap	Action
Low-cost operations	Cost per billing statement	\$0.0813; target is \$0.06	Eliminate microfilming and storage of billing statements
Consistent quality	Percent errors in bill information	0.90%; acceptable	No action
Volume flexibility	Utilization	98%; too high to support rapid increase in volumes	Acquire temporary employees

Extract of Table 1.3 (operations strategy assessment of the billing and payment process), Krajewski [p. 35]

16	IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects. *Unit I | SEE Apr/May 2024 (21IM53), Q2b*	08	3	2

Answer (quoted from the textbook)

Note: The textbook does not discuss IKEA. The activities of operations managers listed in the textbook are given below (eight aspects).

"At the strategic level, operations managers are involved in the development of new capabilities and the maintenance of existing capabilities to best serve the firm's external customers." [p. 40]

"Operations managers design new processes that have strategic implications, and they are deeply involved in the development and organization of supply chains that link external suppliers and external customers to the firm's internal processes." [p. 40]

"Operations managers are often responsible for key performance measures such as cost and quality. These decisions have strategic impact because they affect the processes the firm uses to gain a competitive edge." [p. 40]

"Operations managers are also involved in tactical decisions, including process improvement and performance measurement, managing and planning projects, generating production and staffing plans, managing inventories, and scheduling resources." [p. 40]

Eight aspects: (1) developing and maintaining capabilities; (2) designing new processes; (3) developing and organizing

supply chains; (4) key performance measures such as cost and quality; (5) process improvement and performance measurement; (6) managing and planning projects; (7) generating production and staffing plans; (8) managing inventories and scheduling resources.

"The operations manager's decisions should reflect corporate strategy." [p. 40]

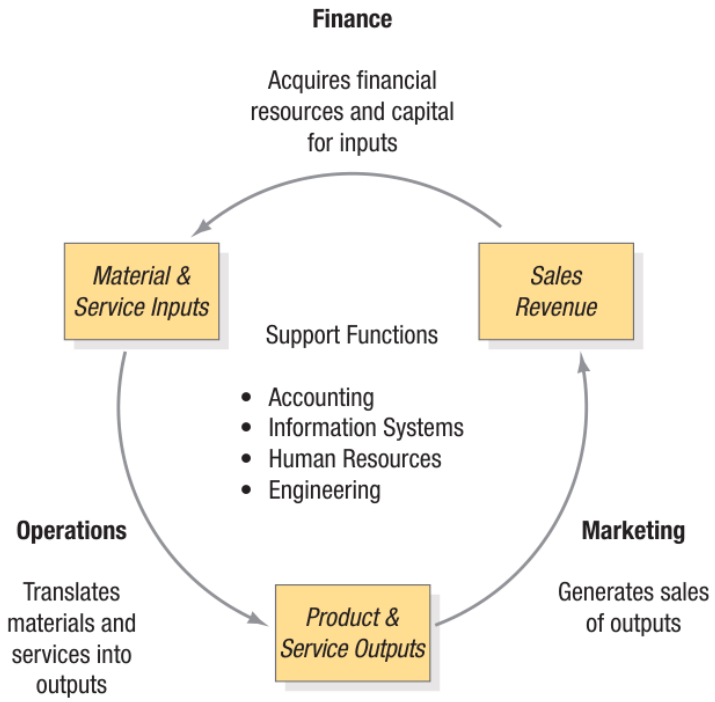
17	Explain the term operations management and identify the three major functional areas of business organizations and briefly describe how they are interrelated. *Unit I | SEE Mar/Apr 2023 (18IM54), Q2a*	08	–	–

Answer (quoted from the textbook)

"Operations management refers to the systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external customers." [p. 22]

Three major functional areas

"Figure 1.1 shows operations as one of the key functions within an organization. The circular relationships in Figure 1.1 highlight the importance of the coordination among the three mainline functions of any business, namely, (1) operations, (2) marketing, and (3) finance. Each function is unique and has its own knowledge and skill areas, primary responsibilities, processes, and decision domains." [p. 23]



▲ **FIGURE 1.1**
Integration between Different Functional Areas of a Business

Figure 1.1: Integration between Different Functional Areas of a Business (reproduced from the textbook)

How they are interrelated

"From an external perspective, finance generates resources, capital, and funds from investors and sales of its goods and services in the marketplace. Based on business strategy, the finance and operations functions then decide how to invest these resources and convert them into physical assets and material inputs. Operations subsequently transforms these material and service inputs into product and service outputs. These outputs must match the characteristics that can be sold in the selected markets by marketing. Marketing is responsible for producing sales revenue of the outputs, which become returns to investors and capital for supporting operations." [p. 23]

"Functions such as accounting, information systems, human resources, and engineering make the firm complete by providing essential information, services, and other managerial support." [p. 23]

"How well these functions work together determines the effectiveness of the organization. Functions should be integrated and should pursue a common strategy." [p. 23]

18	Describe the new trends and challenges in operations management. *Unit I | SEE Mar/Apr 2023 (18IM54), Q2b*	08	–	–

Answer (quoted from the textbook)

"Several trends are currently having a great impact on operations management: productivity improvement; global competition; and ethical, workforce diversity, and environmental issues." [p. 36]

"Accelerating change in the form of information technology, e-commerce, robotics, and the Internet is dramatically affecting the design of new services and products as well as a firm's sales, order fulfillment, and purchasing processes." [p. 36]

1. Productivity improvement

"Improving productivity is a major trend in operations management because all firms face pressures to improve their processes and supply chains so as to compete with their domestic and foreign competitors. Productivity is the value of outputs (services and products) produced divided by the values of input resources (wages, cost of equipment, and so on):" [p. 36]

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$
(Krajewski p. 36)

2. Global competition

"Most businesses realize that, to prosper, they must view customers, suppliers, facility locations, and competitors in global terms." [p. 37]

"Five developments spurred the need for sound global strategies: (1) improved transportation and communications technologies, (2) loosened regulations on financial institutions, (3) increased demand for imported services and goods, (4) reduced import quotas and other international trade barriers due to the formation of regional trading blocks, such as the European Union (EU) and the North American Free Trade Agreement (NAFTA), and (5) comparative cost advantages." [p. 38]

"A firm may have to relinquish proprietary technology if it turns over some of its component manufacturing to offshore suppliers or if suppliers need the firm's technology to achieve desired quality and cost goals. Political risks may also be involved." [p. 38]

3. Ethical, workforce diversity, and environmental issues

"Businesses face more ethical quandaries than ever before, intensified by an increasing global presence and rapid technological change." [p. 39]

"In the past, many people viewed environmental problems, such as toxic waste, poisoned drinking water, poor air quality, and climate change as quality-of-life issues; now, many people and businesses see them as survival issues." [p. 40]

"The challenge is clear: Issues of ethics, workforce diversity, and the environment are becoming part of every manager's job." [p. 40]

19	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff? *Unit I | SEE Jan/Feb 2021 (18IM54), Q2b*	10	–	–

Answer (quoted from the textbook)

Same problem, same figures and same solution as Part B Q3 (the retailer decision tree, Example A.8 of the textbook). (page 12)

20	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. Identify and discuss the factors connecting corporate strategy and key operations management decisions with block diagram. *Unit I | SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2a (identical)*	08	–	–

Answer (quoted from the textbook)

"Developing a customer-driven operations strategy begins with corporate strategy, which, as shown in Figure 1.5, coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. It provides the resources to develop the firm's core competencies and core processes, and it identifies the strategy the firm will employ in international markets. Based on corporate strategy, a market analysis categorizes the firm's customers, identifies their needs, and assesses competitors' strengths. This information is used to develop competitive priorities. These priorities help managers develop the services or products and the processes needed to be competitive in the marketplace." [pp. 27–28]

▶ **FIGURE 1.5**
Connection Between Corporate Strategy and Key Operations Management Decisions

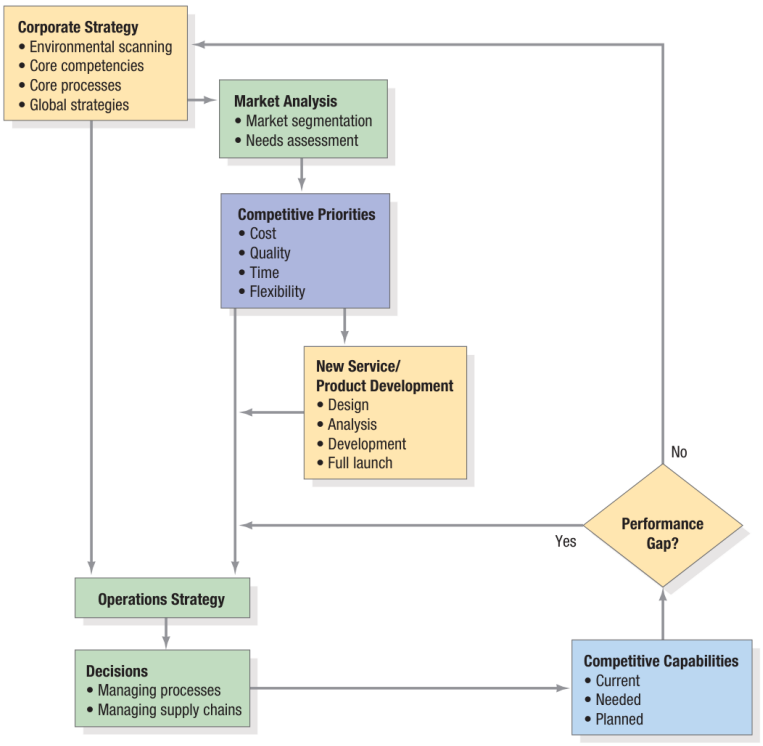


Figure 1.5: Connection Between Corporate Strategy and Key Operations Management Decisions (reproduced from the textbook)

Corporate strategy

"Developing a corporate strategy involves four considerations: (1) monitoring and adjusting to changes in the business environment, (2) identifying and developing the firm's core competencies, (3) developing the firm's core processes, and (4) developing the firm's global strategies." [p. 28]

"Adaptation begins with environmental scanning, the process by which managers monitor trends in the environment (e.g., the industry, the marketplace, and society) for potential opportunities or threats." [p. 28]

"Core competencies are the unique resources and strengths that an organization's management considers when formulating strategy." [p. 29]

"A firm's core competencies should drive its core processes: customer relationship, new service/product development, order fulfillment, and supplier relationship." [p. 29]

Market analysis and competitive priorities

"A market analysis first divides the firm's customers into market segments and then identifies the needs of each segment." [p. 30]

"Competitive priorities are important to the design of existing as well as new services or products, the processes that will deliver them, and the operations strategy that will develop the firm's capabilities to fulfill them. Developing a firm's operations strategy is a continuous process because the firm's capabilities to meet the competitive priorities must be periodically checked and any gaps in performance must be addressed in the operations strategy." [p. 28]

21	Calculate the productivity for the following operations: (i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. (ii) A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labour, \$1,000 for materials, and \$300 for overhead. *Unit I | SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2b (identical)*	08	–	–

Answer (quoted from the textbook)

"Productivity is the value of outputs (services and products) produced divided by the values of input resources (wages, cost of equipment, and so on):" [p. 36]

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$
(Krajewski p. 36)

"Both measures reflect labor productivity, which is an index of the output per person or per hour worked." [p. 36]

"Multifactor productivity is an index of the output provided by more than one of the resources used in production; it may be the value of the output divided by the sum of labor, materials, and overhead costs." [p. 36]

Solution (Example 1.1, textbook)

$$\text{(i) Labor productivity} = \frac{\text{Policies processed}}{\text{Employee hours}} = \frac{600 \text{ policies}}{(3 \text{ employees})(40 \text{ hours/employee})} = 5 \text{ policies/hour}$$
(Example 1.1, Krajewski p. 37)

$$\text{(ii) Multifactor productivity} = \frac{\text{Value of output}}{\text{Labor cost} + \text{Materials cost} + \text{Overhead cost}} = \frac{(400 \text{ units})(\$10/\text{unit})}{\$400 + \$1{,}000 + \$300} = \frac{\$4{,}000}{\$1{,}700} = 2.35$$
(Example 1.1, Krajewski p. 37)

"We want multifactor productivity to be as high as possible. These measures must be compared with performance levels in prior periods and with future goals. If they do not live up to expectations, the process should be investigated for improvement opportunities." [p. 37]

22	Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work. *Unit I | SEE Fast Track Oct 2020 (16IM55/12IM62), Q2a*	06	–	–

Answer (quoted from the textbook)

Definition of operations management and its key terms: see Part B Q1. (page 9)

Block diagram of the process view (Figure 1.2) and how processes work: see Part B Q14. (page 25)

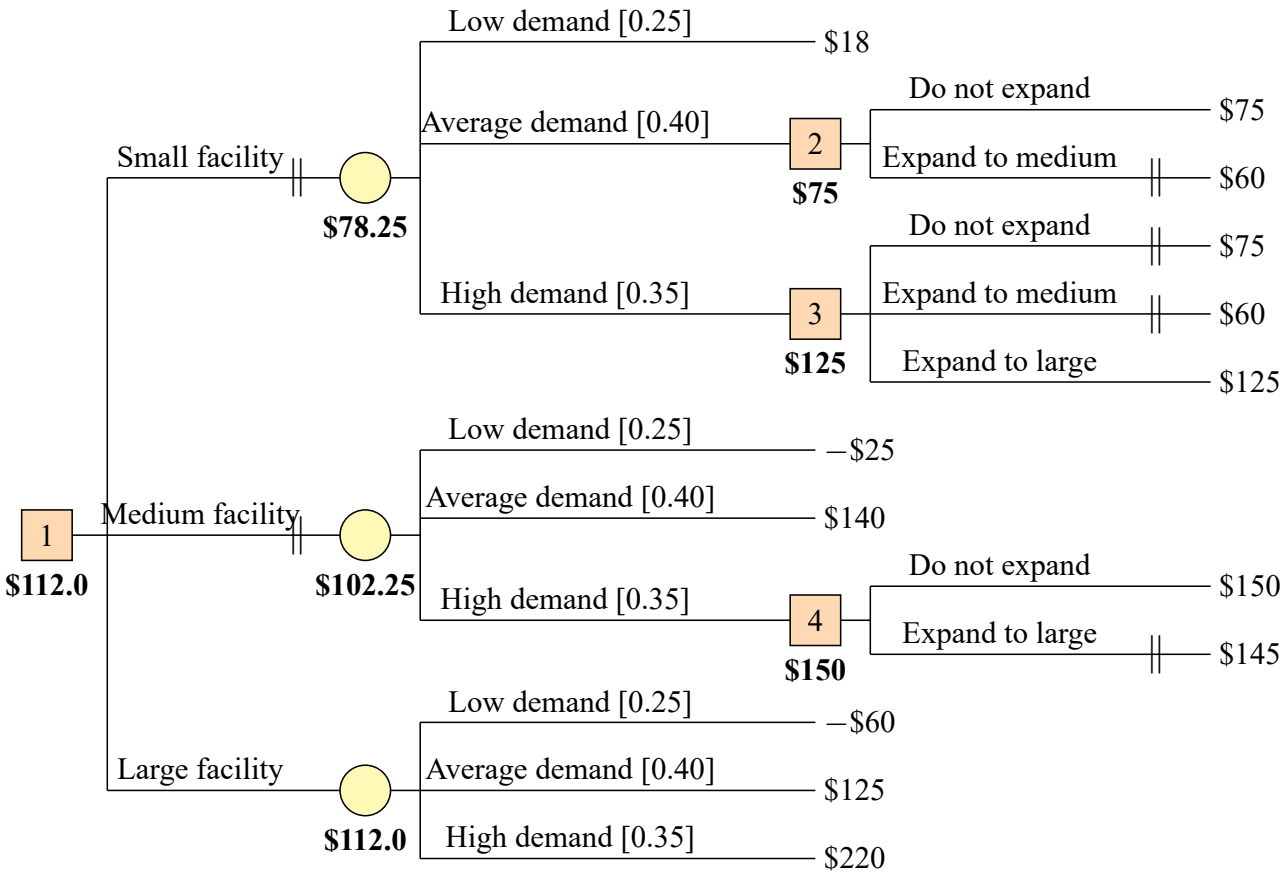
23	A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being 0.25, 0.40 and 0.35 respectively. A small facility is expected to earn an after tax net present value of just $18,000 if demand is low. If demand is average, the small facility is expected to earn $75,000; it can be increased to medium size to earn a net present value of $60,000. If demand is high, the small facility is expected to earn $75,000 and can be expanded to medium size to earn $60,000 or to large size to earn $125,000; a medium sized facility is expected to lose an estimated $25,000 if demand is low and earn $140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of $150,000; it can be expanded to a large size for a net pay off of $145,000. If a large facility is built and demand is high, earnings are expected to be $220,000. If demand is average for the large facility the present value is expected to be $125,000; if demand is low, the facility is expected to lose $60,000. (i) Draw a decision tree for this problem. (ii) What should the management do to achieve the highest expected pay off? *Unit I | SEE Fast Track Oct 2020 (16IM55/12IM62), Q2b*	10	–	–

Answer (quoted from the textbook)

"After drawing a decision tree, we solve it by working from right to left, calculating the expected payoff for each node as follows: 1. For an event node, we multiply the payoff of each event branch by the event's probability. We add these products to get the event node's expected payoff. 2. For a decision node, we pick the alternative that has the best expected payoff." [p. 60]

"A decision tree is a schematic model of alternatives available to the decision maker, along with their possible consequences. The name derives from the tree-like appearance of the model. It consists of a number of square nodes, representing decision points, which are left by branches (which should be read from left to right), representing the alternatives. Branches leaving circular, or chance, nodes represent the events." [p. 59]

(i) Decision tree (payoffs in $000)



Note: The textbook gives no solution diagram for this problem (it is an end-of-chapter problem). The tree above is drawn from the problem data using the conventions of Figures A.4 and A.5: square decision nodes, circular event nodes, probabilities on event branches, "||" marks on pruned branches.

(ii) Analysis, working from right to left

Node 2 (small, average): $\max(75, 60) = 75$

Node 3 (small, high): $\max(75, 60, 125) = 125$

Node 4 (medium, high): $\max(150, 145) = 150$

Small: $0.25(18) + 0.40(75) + 0.35(125) = 4.5 + 30 + 43.75 = 78.25$

Medium: $0.25(-25) + 0.40(140) + 0.35(150) = -6.25 + 56 + 52.5 = 102.25$

Large: $0.25(-60) + 0.40(125) + 0.35(220) = -15 + 50 + 77 = 112.0$

Working (textbook method applied to the numbers of the question)

"For a decision node, we pick the alternative that has the best expected payoff." [p. 60]

Answer: Build the large facility: expected payoff $112,000 (medium $102,250; small $78,250).

24	How does the process view and the supply chain view of operations management help us understand the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management. *Unit I | SEE Fast Track Jul 2019 (16IM55/12IM62), Q2a*	08	–	–

Answer (quoted from the textbook)

Process view

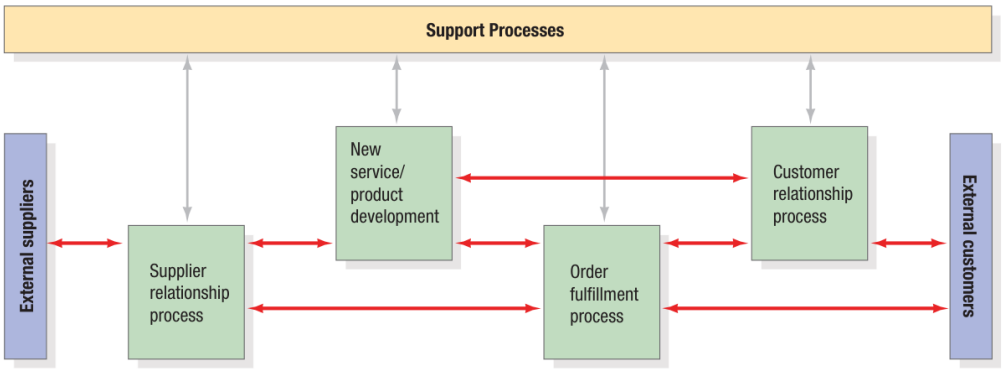
"The reason is that a process view of the firm provides a much more relevant picture of the way firms actually work. Departments typically have their own set of objectives, a set of resources with capabilities to achieve those objectives, and managers and employees responsible for performance." [p. 24]

"The concept of a process, however, can be much broader. A process can have its own set of objectives, involve a work flow that cuts across departmental boundaries, and require resources from several departments." [p. 24]

"Therefore, operations management is relevant and important for all students, regardless of major, because all departments have processes that must be managed effectively to gain a competitive advantage." [p. 24]

Supply chain view

"Most services or products are produced through a series of interrelated business activities. Each activity in a process should add value to the preceding activities; waste and unnecessary cost should be eliminated. Our process view of a firm is helpful for understanding how services or products are produced and why cross-functional coordination is important, but it does not shed any light on the strategic benefits of the processes. The missing strategic insight is that processes must add value for customers throughout the supply chain. The concept of supply chains reinforces the link between processes and performance, which includes a firm's internal processes as well as those of its external customers and suppliers." [p. 26]



▲ **FIGURE 1.4**
Supply Chain Linkages Showing Work and Information Flows

Figure 1.4: Supply Chain Linkages Showing Work and Information Flows (reproduced from the textbook)

"A core process is a set of activities that delivers value to external customers." [p. 27]

"Employees in the supplier relationship process select the suppliers of services, materials, and information and facilitate the timely and efficient flow of these items into the firm." [p. 27]

"Employees in the new service/product development process design and develop new services or products." [p. 27]

"The order fulfillment process includes the activities required to produce and deliver the service or product to the external customer." [p. 27]

"Employees involved in the customer relationship process identify, attract, and build relationships with external customers, and facilitate the placement of orders by customers." [p. 27]

"A support process provides vital resources and inputs to the core processes and is essential to the management of the business." [p. 27]

Operations as a strategic function

"Operations strategy is the linchpin that brings these processes together to form supply chains that extend beyond the walls of the firm, encompassing suppliers as well as customers." [p. 27]

"Operations managers are often responsible for key performance measures such as cost and quality. These decisions have strategic impact because they affect the processes the firm uses to gain a competitive edge." [p. 40]

25	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before commercializing the device, the owner wants reasonable assurance of success. Variable costs are \$7 per unit produced and sold. Fixed costs are about \$500,000 per year. (i) If the price is reduced to \$15 and forecasted sales for the first year are 10,000 units, what would be the product's total contribution to profits in the first year? (ii) If the selling price is set at \$25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches. *Unit I | SEE Fast Track Jul 2019 (16IM55/12IM62), Q2b*	08	–	–

Answer (quoted from the textbook)

"The break-even quantity is the volume at which total revenues equal total costs." [p. 51]

$$\text{Total cost} = F + cQ, \qquad \text{Total revenue} = pQ, \qquad Q = \frac{F}{p - c}$$

(Krajewski p. 52)

(i) Contribution at a price of \$15 and 10,000 units

$$\text{Total profit contribution} = pQ - (F + cQ)$$

(Krajewski p. 65)

$$15(10{,}000) - [500{,}000 + 7(10{,}000)] = 150{,}000 - 570{,}000 = -\$420{,}000$$

Working (textbook method applied to the numbers of the question)

Answer: Contribution = \$420,000 loss in the first year.

(ii) Break-even at a price of \$25

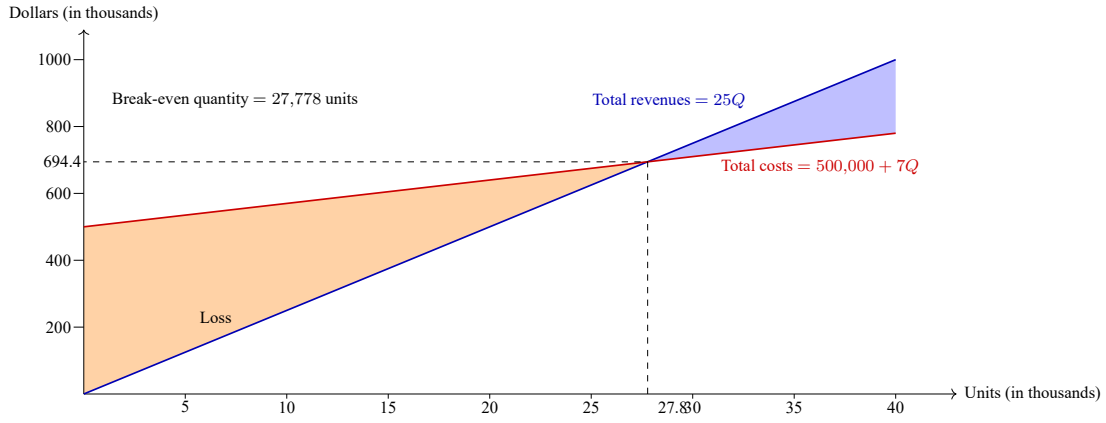
Algebraic approach

$$Q = \frac{F}{p - c} = \frac{500{,}000}{25 - 7} = 27{,}777.8 \approx 27{,}778 \text{ units}$$

Working (textbook method applied to the numbers of the question)

Graphic approach

"Because both costs and revenues are linear relationships, the break-even quantity is where the total revenue line crosses the total cost line." [p. 52]



Lines drawn: total revenue = 25Q, passing through (0, 0) and (40,000 units, $1,000,000); total cost = 500,000 + 7Q, passing through (0, $500,000) and (40,000 units, $780,000). They intersect at Q = 27,778 units.

Answer: Break-even quantity = 27,778 units.

Note: The textbook gives this problem with fixed costs of $56,000 (Solved Problem 4, see Part B Q26). The method quoted above is the textbook's; the figure and arithmetic apply it to the $500,000 fixed cost of this question, and the chart follows the style of the textbook's Figure A.7.

26	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Variable costs are estimated at $7 per unit produced and sold. Fixed costs are about $56,000 per year. (i) If the selling price is set at $25, how many units must be produced and sold to break even? (ii) Forecasted sales for the first year are 10,000 units if the price is reduced to $15. With this pricing strategy, what would be the product's total contribution to profits in the first year? *Unit I | SEE Feb/Mar 2026 (IM352IA), Q2a*	08	3	1

Answer (quoted from the textbook)

Solution (Solved Problem 4, textbook)

"The break-even quantity is the volume at which total revenues equal total costs." [p. 51]

(i) Break-even quantity at $25: algebraic approach

$$Q = \frac{F}{p - c} = \frac{56{,}000}{25 - 7} = 3{,}111 \text{ units}$$

(Solved Problem 4, Krajewski p. 65)

Graphic approach (Figure A.7, textbook)

Two lines are drawn: Total revenue = 25Q and Total cost = 56,000 + 7Q. The two lines intersect at Q = 3,111 units, the break-even quantity.

◀ FIGURE A.7

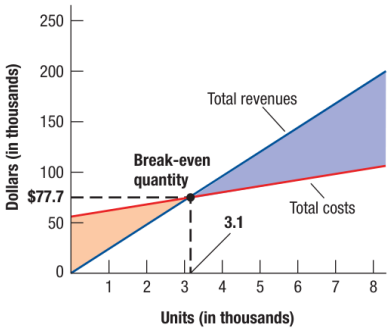


Figure A.7: break-even chart (reproduced from the textbook)

(ii) Contribution at $15 and 10,000 units

$$\text{Total profit contribution} = \text{Total revenue} - \text{Total cost} = pQ - (F + cQ) = 15(10{,}000) - [56{,}000 + 7(10{,}000)] = \$24{,}000$$

(Solved Problem 4, Krajewski p. 65)

27	With suitable example, distinguish between manufacturing processes and service processes. *Unit I | SEE Feb/Mar 2026 (IM352IA), Q2b*	08	2	1

Answer (quoted from the textbook)

Same question as Part B Q2. (page 10)

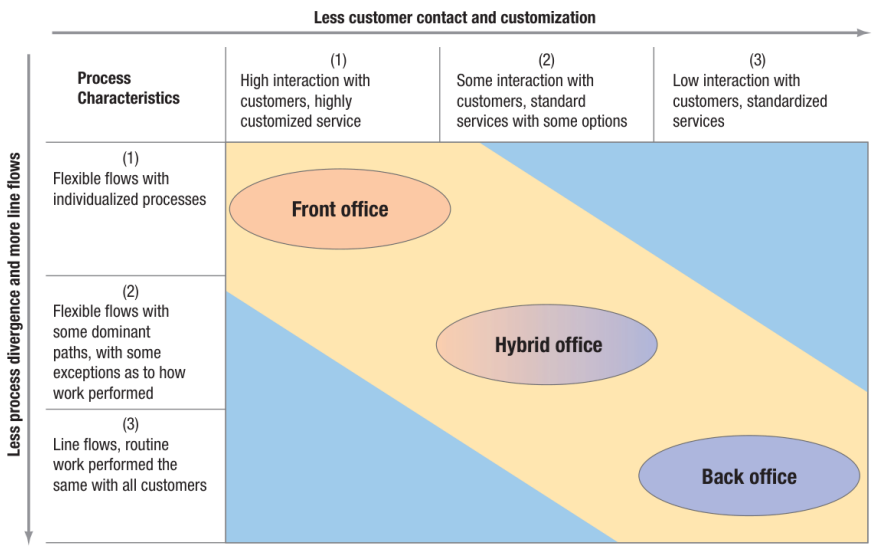
28	Compare process structures in services and manufacturing. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q3a*	08	4	3

Answer (quoted from the textbook)

Services: customer-contact matrix

"The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics. The matrix is the starting point for evaluating and improving a process." [p. 113]

"The manager has three process structures, which form a continuum, to choose from: (1) front office, (2) hybrid office, and (3) back office." [p. 114]



◀ **FIGURE 3.2**
Customer-Contact Matrix for Service Processes

Figure 3.2: Customer-Contact Matrix for Service Processes (reproduced from the textbook)

"A front-office process has high customer contact where the service provider interacts directly with the internal or external customer." [p. 114]

"A hybrid-office process has moderate levels of customer contact and standard services, with some options available from which the customer chooses." [p. 114]

"A back-office process has low customer contact and little service customization." [p. 114]

Manufacturing: product-process matrix

"The product-process matrix, shown in Figure 3.3, brings together three elements: (1) volume, (2) product customization, and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process itself." [p. 114]

"The manager has four process choices, which form a continuum, to choose from: (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process." [p. 115]

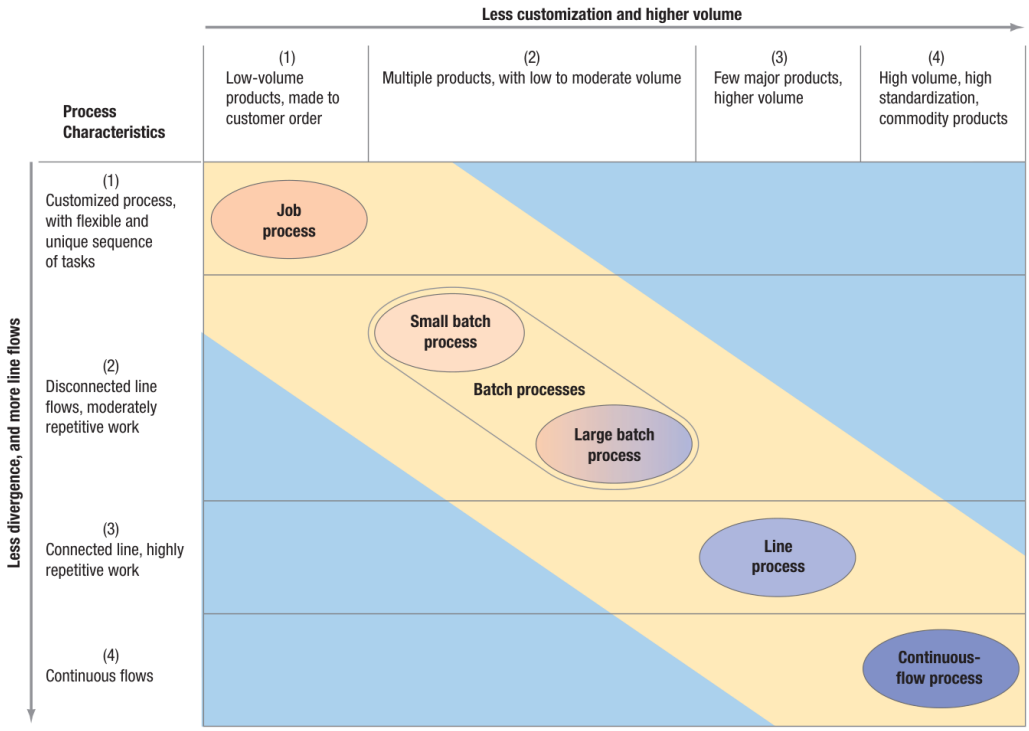


Figure 3.3: Product-Process Matrix for Manufacturing Processes (reproduced from the textbook)

Comparison (each cell quoted from the textbook)

Basis	Services	Manufacturing
Matrix and its elements	The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics.	The product-process matrix, shown in Figure 3.3, brings together three elements: (1) volume, (2) product customization, and (3) process characteristics.
Process structures	The manager has three process structures, which form a continuum, to choose from: (1) front office, (2) hybrid office, and (3) back office.	The manager has four process choices, which form a continuum, to choose from: (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process.
Customer contact and inventory	Customer contact is the extent to which the customer is present, is actively involved, and receives personal attention during the service process.	Customer contact, a primary feature of the customer-contact matrix for services, normally is not a consideration for manufacturing processes

[pp. 113–115]

"Strategies for manufacturing processes differ from those in services, not only because of low customer contact and involvement, but also because of the ability to use inventories." [p. 116]

29	Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q3b*	08	4	3

Answer (quoted from the textbook)

"Capacity is the maximum rate of output of a process or a system." [p. 222]

"Long-term capacity plans deal with investments in new facilities and equipment at the organizational level, and require top management participation and approval because they are not easily reversed. These plans cover at least two years into the future, but construction lead times can sometimes be longer and result in longer planning time horizons." [p. 223]

"Operations managers must examine three dimensions of capacity strategy before making capacity decisions: (1) sizing capacity cushions, (2) timing and sizing expansion, and (3) linking process capacity and other operating decisions." [p. 225]

Sizing capacity cushions

"The capacity cushion is the amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent." [p. 225]

$$\text{Capacity cushion} = 100(\%) - \text{Average utilization rate}(\%)$$

(Krajewski p. 225)

Timing and sizing of expansion

"The second issue of capacity strategy concerns when to adjust capacity levels and by how much." [p. 226]

"Figure 6.2 illustrates two extreme strategies for expanding capacity: the expansionist strategy, which involves large, infrequent jumps in capacity, and the wait-and-see strategy, which involves smaller, more frequent jumps." [p. 226]

FIGURE 6.2 ▶
Two Capacity Strategies

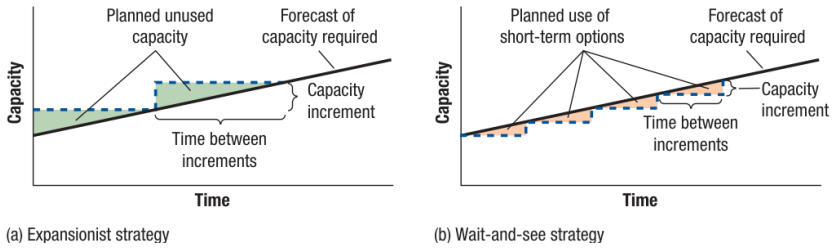


Figure 6.2: Two Capacity Strategies (reproduced from the textbook)

"The timing and sizing of expansion are related; that is, if demand is increasing and the time between increments increases, the size of the increments must also increase. The expansionist strategy, which stays ahead of demand, minimizes the chance of sales lost to insufficient capacity. The wait-and-see strategy lags behind demand. To meet any shortfalls, it relies on short-term options, such as use of overtime, temporary workers, subcontractors, stockouts, and the postponement of preventive maintenance on equipment." [p. 226]

"Several factors favor the expansionist strategy. Expansion can result in economies of scale and a faster rate of learning, thus helping a firm reduce its costs and compete on price." [p. 226]

"The conservative wait-and-see strategy is to expand in smaller increments, such as by renovating existing facilities rather than building new ones. Because the wait-and-see strategy follows demand, it reduces the risks of overexpansion based on overly optimistic demand forecasts, obsolete technology, or inaccurate assumptions regarding the competition." [p. 227]

"However, this strategy has its own risks, such as being preempted by a competitor or being unable to respond if demand is unexpectedly high." [p. 227]

"An intermediate strategy could be to follow the leader, expanding when others do." [p. 227]

Linking capacity and other decisions

"Capacity decisions should be closely linked to processes and supply chains throughout the organization." [p. 227]

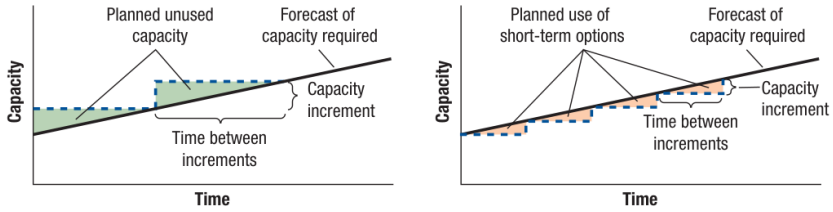
30	Explain timing and sizing strategies in capacity planning with diagrams. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q4a*	08	4	3

Answer (quoted from the textbook)

"The second issue of capacity strategy concerns when to adjust capacity levels and by how much." [p. 226]

"Figure 6.2 illustrates two extreme strategies for expanding capacity: the expansionist strategy, which involves large, infrequent jumps in capacity, and the wait-and-see strategy, which involves smaller, more frequent jumps." [p. 226]

FIGURE 6.2 ▶
Two Capacity Strategies



(a) Expansionist strategy (b) Wait-and-see strategy

Figure 6.2: Two Capacity Strategies (reproduced from the textbook)

(a) Expansionist strategy

"The expansionist strategy, which stays ahead of demand, minimizes the chance of sales lost to insufficient capacity." [p. 226]

"By making a large capacity expansion or announcing that one is imminent, the firm can preempt the expansion of other firms." [p. 226]

(b) Wait-and-see strategy

"The wait-and-see strategy lags behind demand. To meet any shortfalls, it relies on short-term options, such as use of overtime, temporary workers, subcontractors, stockouts, and the postponement of preventive maintenance on equipment." [p. 226]

"Because the wait-and-see strategy follows demand, it reduces the risks of overexpansion based on overly optimistic demand forecasts, obsolete technology, or inaccurate assumptions regarding the competition." [p. 227]

Timing and sizing are related

"The timing and sizing of expansion are related; that is, if demand is increasing and the time between increments increases, the size of the increments must also increase." [p. 226]

"Management may choose one of these two strategies or one of the many between these extremes." [p. 227]

31	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability? *Unit II | SEE Feb/Mar 2026 (IM352IA), Q4b*	08	2	2

Answer (quoted from the textbook)

eBay: accommodating customer-introduced variability

"How much should customers be involved in a process, so as to provide timely delivery and consistent quality, and at sustainable cost? Various ways are available—some accommodate customer-introduced variability and some reduce it. eBay illustrates one way to accommodate variability." [p. 109]

"As an online auction house, eBay has high volume and request variability. Its customers do not want service at the same time or at times necessarily convenient to the company. They have request variability, seeking to buy and sell an endless number of items. They also have variability in customer capability, some with considerable Internet experience and some needing more handholding. Such variability would greatly complicate workforce scheduling if eBay required its employees to conduct all of its processes." [p. 109]

"This relatively small workforce is possible in the face of customer-induced variability because its customers perform virtually all of the selling and buying processes through the eBay Web site. When the customer is responsible for

much of the work, the right labor is provided at the right moment." [p. 110]

McDonald's

"McDonald's standardizes work methods, staff training processes, and procurement of raw materials to achieve the same consistent product and process quality from one store to the next." [p. 32]

"For example, the process of cooking a hamburger at a McDonald's is a manufacturing process because it changes the material's physical properties (dimension 1), as is the process of assembling the hamburger with the bun (dimension 5)." [p. 26]

Link to customer-introduced variability

"Such customer contact can introduce considerable variability, disrupting carefully designed production processes. Costs and quality can be adversely affected." [p. 109]

Note: The chapter describes eBay's process strategy in full but gives McDonald's only the statements quoted above (Chapters 1 and 3 of the supplied units); it does not describe in detail how McDonald's reduces customer-introduced variability.

32	Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q3a*	10	3	2

Answer (quoted from the textbook)

"Make-to-order, assemble-to-order, and make-to-stock strategies are three approaches to inventory that should be coordinated with process choice." [p. 116]

Note: The textbook names three inventory strategies (below); it also describes mass production and mass customization as related terms (quoted at the end).

1. Make-to-order

"Manufacturers that make products to customer specifications in low volumes tend to use the make-to-order strategy, coupling it with job or small batch processes. It is a more complex process than assembling a final product from standard components. This strategy provides a high degree of customization and typically uses job or small batch processes. The processes have high divergence. Specialized medical equipment, castings, and expensive homes are suited to the make-to-order strategy." [p. 116]

2. Assemble-to-order

"The assemble-to-order strategy is an approach to producing a wide variety of products from relatively few subassemblies and components after the customer orders are received. Typical competitive priorities are variety and fast delivery times. The assemble-to-order strategy often involves a line process for assembly and a batch process for fabrication." [p. 116]

"Stocking finished products would be economically prohibitive because the numerous possible options make forecasting relatively inaccurate. Thus, the principle of postponement is applied, whereby the final activities in the provision of a product are delayed until the orders are received. The assemble-to-order strategy is also linked to mass customization, where highly divergent processes generate a wide variety of customized products at reasonably low costs." [pp. 116–117]

3. Make-to-stock

"Manufacturing firms that hold items in stock for immediate delivery, thereby minimizing customer delivery times, use a make-to-stock strategy. This strategy is feasible for standardized products with high volumes and reasonably accurate forecasts. It is the inventory strategy of choice for line or continuous-flow processes. Examples of products produced with a make-to-stock strategy include garden tools, electronic components, soft drinks, and chemicals." [p. 117]

"Combining a line process with the make-to-stock strategy is sometimes called mass production." [p. 117]

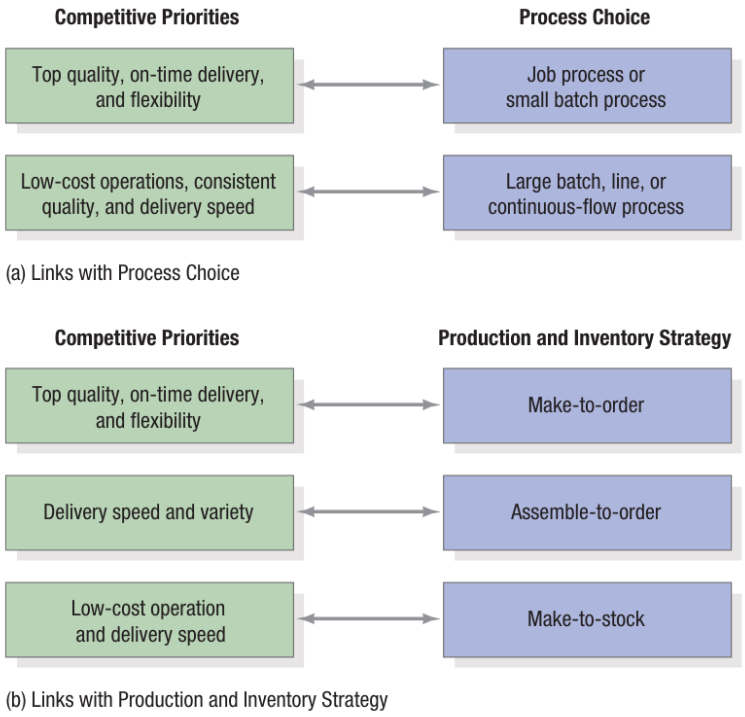


Figure 3.9: Links of Competitive Priorities with Manufacturing Strategy (reproduced from the textbook)

33	Provide a note on the three elements in a customer-contract matrix in service processes. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q3b*	06	2	2

Answer (quoted from the textbook)

Note: The question says "customer-contract matrix"; the textbook's term is the customer-contact matrix.

"The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics. The matrix is the starting point for evaluating and improving a process." [p. 113]

(1) Degree of customer contact

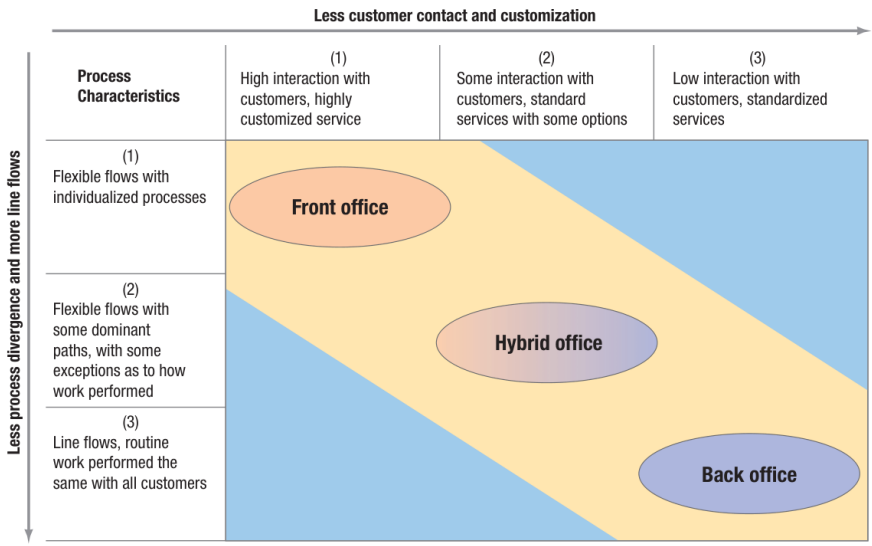
"Customer contact is the extent to which the customer is present, is actively involved, and receives personal attention during the service process." [p. 113]

(2) Customization

"The horizontal dimension of the matrix represents the service provided to the customer in terms of customer contact and competitive priorities. A key competitive priority is how much customization is needed. Positions on the left side of the matrix represent high customer contact and highly customized services." [p. 113]

(3) Process characteristics

"The vertical dimension of the customer-contact matrix deals with two characteristics of the process itself: (1) process divergence and (2) flow. Each process can be analyzed on these two dimensions." [p. 113]



◀ **FIGURE 3.2**
Customer-Contact Matrix for Service Processes

Figure 3.2: Customer-Contact Matrix for Service Processes (reproduced from the textbook)

34	Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q4a*	10	3	2

Answer (quoted from the textbook)

"Operations managers must examine three dimensions of capacity strategy before making capacity decisions: (1) sizing capacity cushions, (2) timing and sizing expansion, and (3) linking process capacity and other operating decisions." [p. 225]

(1) Sizing capacity cushions

"The capacity cushion is the amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent." [p. 225]

"The appropriate size of the cushion varies by industry. In the capital-intensive paper industry, where machines can cost hundreds of millions of dollars each, cushions well under 10 percent are preferred. The less capital-intensive hotel industry breaks even with a 60 to 70 percent utilization (40 to 30 percent cushion), and begins to suffer customer-service problems when the cushion drops to 20 percent. The more capital-intensive cruise ship industry prefers cushions as small as 5 percent." [p. 225]

"Prompt customer service requires supermarkets to maintain a capacity cushion large enough to handle peak demand." [p. 225]

(2) Timing and sizing expansion

"The second issue of capacity strategy concerns when to adjust capacity levels and by how much." [p. 226]

"General Motors decided to increase production capacity of the four-seat series hybrid car Chevrolet Volt from 30,000 units to 45,000 units in 2012 because of strong public interest." [p. 226]

"Managerial Practice 6.1 illustrates the use of expansionist strategy in the ethanol industry." [p. 226]

(3) Linking capacity and other decisions

"Capacity decisions should be closely linked to processes and supply chains throughout the organization. When managers make decisions about designing processes, determining degree of resource flexibility and inventory, and locating facilities, they must consider its impact on capacity cushions." [pp. 227–228]

"For example, capacity cushions for a process can be lowered if less emphasis is placed on fast deliveries (competitive priorities), yield losses (quality) drop, or if investment in capital-intensive equipment increases or worker flexibility increases (process design)." [p. 228]

35	Enumerate the four steps of systematic approach to long term capacity decisions. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q4b*	06	2	2

Answer (quoted from the textbook)

"Although each situation is somewhat different, a four-step procedure generally can help managers make sound capacity decisions." [p. 228]

"1. Estimate future capacity requirements. 2. Identify gaps by comparing requirements with available capacity. 3. Develop alternative plans for reducing the gaps. 4. Evaluate each alternative, both qualitatively and quantitatively, and make a final choice." [p. 228]

Step 1

"A process's capacity requirement is what its capacity should be for some future time period to meet the demand of the firm's customers (external or internal), given the firm's desired capacity cushion." [p. 228]

Step 2

"A capacity gap is any difference (positive or negative) between projected capacity requirements (M) and current capacity." [p. 230]

Step 3

"The next step is to develop alternative plans to cope with projected gaps. One alternative, called the base case, is to do nothing and simply lose orders from any demand that exceeds current capacity or incur costs because capacity is too large." [p. 230]

Step 4

"In this final step, the manager evaluates each alternative, both qualitatively and quantitatively." [p. 230]

36	List and explain operations process characteristics. Also briefly discuss the implication of four characteristics of operations processes. *Unit II | SEE Apr/May 2024 (21IM53), Q3a*	08	3	2

Answer (quoted from the textbook)

Process characteristics

"The vertical dimension of the customer-contact matrix deals with two characteristics of the process itself: (1) process divergence and (2) flow." [p. 113]

"Process divergence is the extent to which the process is highly customized with considerable latitude as to how its tasks are performed." [p. 113]

"A flexible flow means that the customers, materials, or information move in diverse ways, with the path of one customer or job often crisscrossing the path that the next one takes." [p. 113]

"A line flow means that the customers, materials, or information move linearly from one operation to the next, according to a fixed sequence." [p. 114]

The four major process decisions

"Whether dealing with processes for offices, service providers, or manufacturers, operations managers must consider four common process decisions." [p. 112]

"Process structure determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A layout, which is the physical arrangement of operations created from the various processes, puts these decisions into tangible form." [p. 112]

"Customer involvement reflects the ways in which customers become part of the process and the extent of their participation." [p. 112]

"Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions." [p. 112]

"Capital intensity is the mix of equipment and human skills in a process. The greater the relative cost of equipment, the greater is the capital intensity." [p. 112]

Implications

"The key to successful process decisions is to make choices that fit the situation and that make sense together. They should not work at cross-purposes, with one process optimized at the expense of other processes." [p. 110]

The implications of high customer contact (services) and high volume (manufacturing) for each of the four decisions are quoted in Part B Q6. (page 16)

37	How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization? *Unit II | SEE Apr/May 2024 (21IM53), Q3b*	08	3	2

Answer (quoted from the textbook)

"A good process strategy for a service process depends first and foremost on the type and amount of customer contact. Customer contact is the extent to which the customer is present, is actively involved, and receives personal attention during the service process." [pp. 112–113]

TABLE 3.3 DIMENSIONS OF CUSTOMER CONTACT IN SERVICE PROCESSES

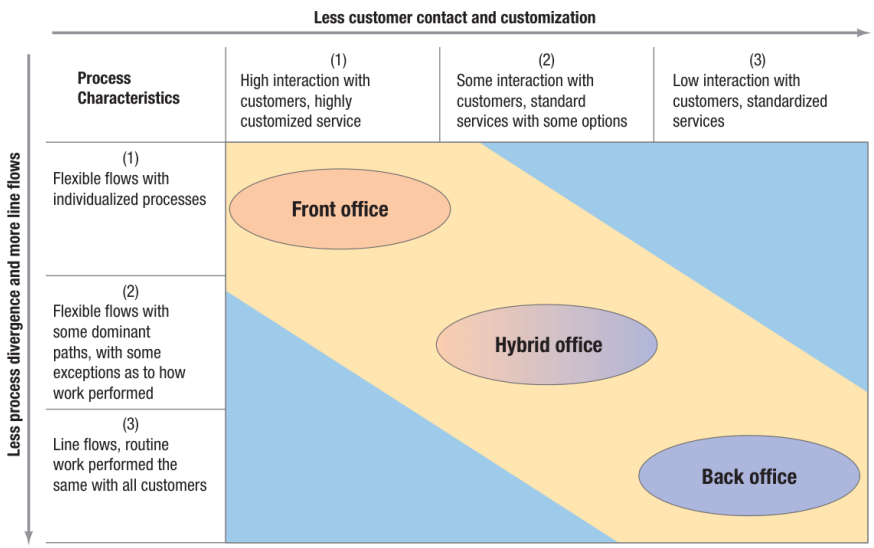
Dimension	High Contact	Low Contact
Physical presence	Present	Absent
What is processed	People	Possessions or information
Contact intensity	Active, visible	Passive, out of sight
Personal attention	Personal	Impersonal
Method of delivery	Face-to-face	Regular mail or e-mail

Table 3.3: Dimensions of Customer Contact in Service Processes (reproduced from the textbook)

"The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics." [p. 113]

"A key competitive priority is how much customization is needed. Positions on the left side of the matrix represent high customer contact and highly customized services. The customer is more likely to be present and active. The process is more likely to be visible to the customer, who receives more personal attention. The right side of the matrix represents low customer contact, passive involvement, less personalized attention, and a process out of the customer's sight." [p. 113]

"Process divergence is the extent to which the process is highly customized with considerable latitude as to how its tasks are performed." [p. 113]



◀ **FIGURE 3.2**
Customer-Contact Matrix for Service Processes

Figure 3.2: Customer-Contact Matrix for Service Processes (reproduced from the textbook)

Resulting structures

"A front-office process has high customer contact where the service provider interacts directly with the internal or external customer. Because of the customization of the service and variety of service options, many of the steps in it have considerable divergence. Work flows are flexible, and they vary from one customer to the next." [p. 114]

"A hybrid-office process has moderate levels of customer contact and standard services, with some options available from which the customer chooses. The work flow progresses from one workstation to the next, with some dominant paths apparent." [p. 114]

"A back-office process has low customer contact and little service customization. The work is standardized and routine, with line flows from one service provider to the next until the service is completed." [p. 114]

38	Explain the concept of economies of scale when planning capacity for long term implementation. *Unit II | SEE Apr/May 2024 (21IM53), Q4a*	08	2	2

Answer (quoted from the textbook)

Economies of scale, its four reasons, diseconomies of scale and Figure 6.1 are quoted in Part B Q8. (page 18)

"Deciding on the best level of capacity involves consideration for the efficiency of the operations. A concept known as economies of scale states that the average unit cost of a service or good can be reduced by increasing its output rate." [p. 224]

"Figure 6.1 does not mean that the optimal size for all hospitals is 500 beds. Optimal size depends on the number of patients per week to be served." [p. 225]

39	The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix. *Unit II | SEE Apr/May 2024 (21IM53), Q4b*	08	3	3

Answer (quoted from the textbook)

"The product-process matrix, shown in Figure 3.3, brings together three elements: (1) volume, (2) product customization, and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process itself." [p. 114]

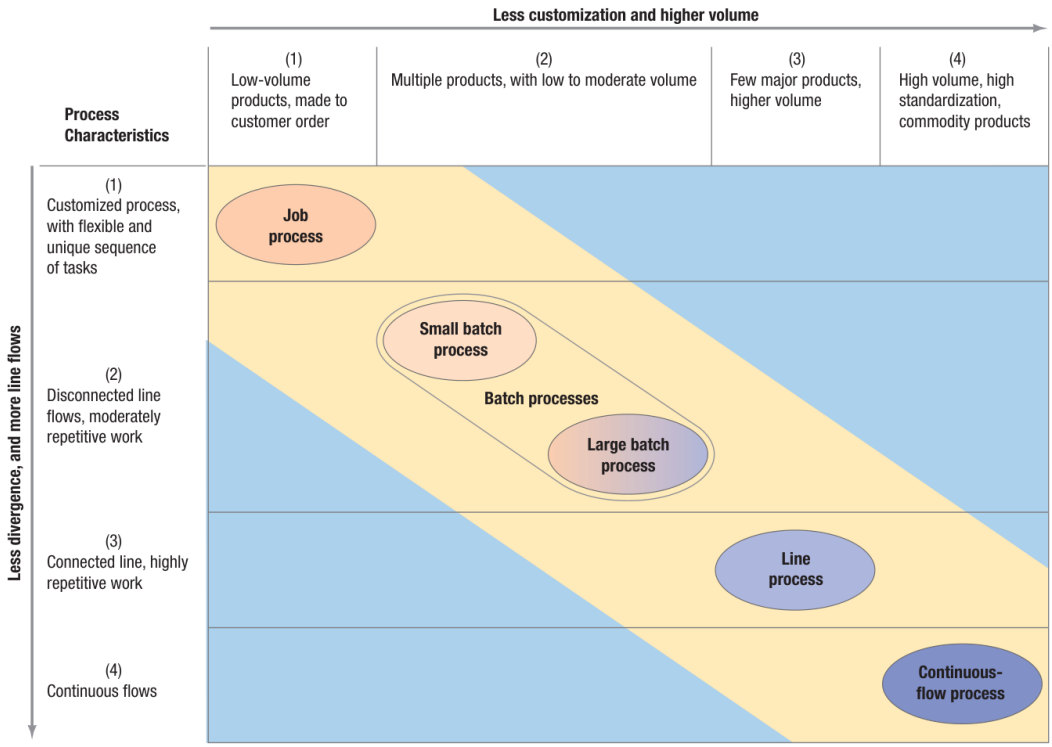


Figure 3.3: Product-Process Matrix for Manufacturing Processes (reproduced from the textbook)

"The fundamental message in Figure 3.3 is that the best choice for a manufacturing process depends on the volume and degree of customization required of the process." [p. 115]

"As with the customer-contact matrix, it is unlikely that a manufacturing process can be a top performer if its position is too far from the diagonal." [p. 115]

Justification: the four process choices

"A job process creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low. The workforce and equipment are flexible to handle considerable task divergence. Companies choosing job processes often bid for work. Typically, they make products to order and do not produce them ahead of time. Each new order is handled as a single unit—as a job. Examples are machining a metal casting for a customized order or producing customized cabinets." [p. 115]

"A batch process has average or moderate volumes, but process divergence is still too great to warrant dedicating a separate process for each product. The process flow is flexible, but more dominant paths emerge than at a job process, and some segments of the process have a line flow. Examples of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment." [p. 116]

"A line process lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products. Divergence is minimal in the process

or line flows, and little inventory is held between the processing steps. Each step performs the same process over and over, with little variability in the products manufactured. Production and material handling equipment is specialized. Products created by a line process include the assembly of computers, automobiles, appliances, and toys." [p. 116]

"A continuous flow process is the extreme end of high-volume standardized production, with rigid line flows. Process divergence is negligible. Its name derives from the way materials move through the process. Usually, one primary material (such as a liquid, a gas, or a powder) moves without stopping through the process. A continuous-flow process differs from a line process in one important respect: Materials (be they undifferentiated or discrete) flow through the process without stopping until the whole batch is finished. The time span can be several shifts or even several months. Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food" [p. 116]

40	Describe the four basic process strategies. *Unit II | SEE Mar/Apr 2023 (18IM54), Q3a*	08	–	–

Answer (quoted from the textbook)

"We begin by defining four basic process decisions: (1) process structure (including layout), (2) customer involvement, (3) resource flexibility, and (4) capital intensity." [p. 110]

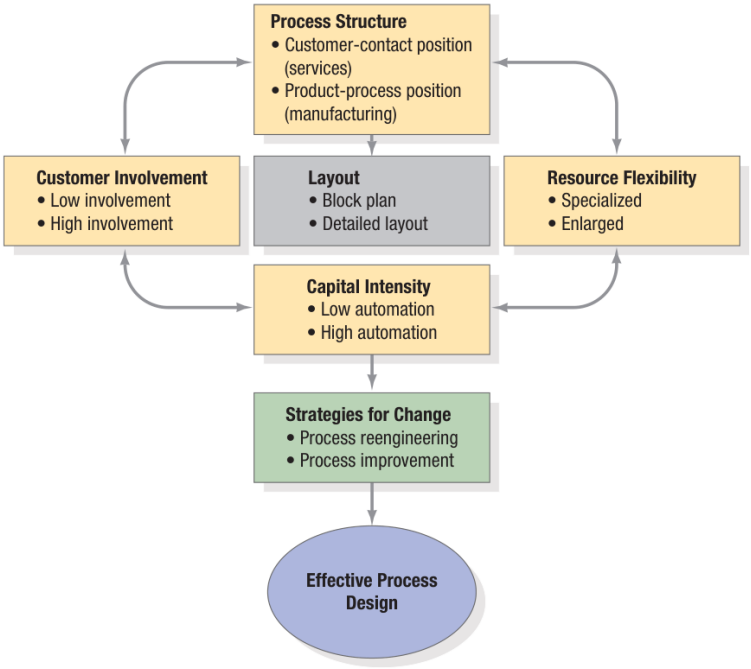


Figure 3.1: Major Decisions for Effective Processes (reproduced from the textbook)

"Process structure determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A layout, which is the physical arrangement of operations created from the various processes, puts these decisions into tangible form." [p. 112]

"Customer involvement reflects the ways in which customers become part of the process and the extent of their participation." [p. 112]

"Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions." [p. 112]

"Capital intensity is the mix of equipment and human skills in a process. The greater the relative cost of equipment, the greater is the capital intensity." [p. 112]

"We conclude with two basic change strategies for analyzing and modifying processes: (1) process reengineering and (2) process improvement." [p. 110]

41	List the different tools for process analysis and design and explain them. *Unit II | SEE Mar/Apr 2023 (18IM54), Q3b*	08	–	–

Answer (quoted from the textbook)

"We discuss the tools that help managers analyze processes, and we reveal the methods firms use to measure process performance and quality." [p. 41]

Note: Tool definitions below are taken from the textbook Glossary. The textbook index places these tools in Chapter 4 (pp. 139–160: flowcharts, swim lane flowcharts, service blueprints, process charts, time study, work sampling, histograms, Pareto charts, scatter diagrams, cause-and-effect diagrams, brainstorming, benchmarking); the chapter text itself is not in the supplied units.

Tool	**Definition (Glossary)**
Flowchart	A diagram that traces the flow of information, customers, equipment, or materials through the various steps of a process.
Swim lane flowchart	A visual representation that groups functional areas responsible for different sub-processes into lanes.
Service blueprint	A special flowchart of a service process that shows which steps have high customer contact.
Process chart	An organized way of documenting all the activities performed by a person or group of people, at a workstation, with a customer, or on materials.
Time study	A work measurement method using a trained analyst to perform four basic steps in setting a time standard for a job or process: selecting the work elements (or nested processes) within the process to be studied, timing the elements, determining the sample size, and setting the final standard.
Work sampling method	A process that estimates the proportion of time spent by people or machines on different activities, based on observations randomized over time.
Histogram	A summarization of data measured on a continuous scale, showing the frequency distribution of some process failure (in statistical terms, the central tendency and dispersion of the data).
Pareto chart	A bar chart on which factors are plotted along the horizontal axis in decreasing order of frequency.
Scatter diagram	A plot of two variables showing whether they are related.
Cause-and-effect diagram	A diagram that relates a key performance problem to its potential causes.
Brainstorming	Letting a group of people, knowledgeable about the process, propose ideas for change by saying whatever comes to mind.
Benchmarking	A systematic procedure that measures a firm's processes, services, and products against those of industry leaders.
Metrics	Performance measures that are established for a process and the steps within it.

[Glossary, pp. 639–652]

42	Define strategic fit. Describe the elements of strategic fit. *Unit II | SEE Mar/Apr 2023 (18IM54), Q4a*	08	–	–

Answer (quoted from the textbook)

Definition

"The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective." [p. 124]

"A more effective process is one that matches key process characteristics and has a close strategic fit." [p. 110]

Elements of strategic fit

"The concepts that we develop around these four decisions establish a framework within which we can address the appropriate process design in every situation." [p. 112]

"Process structure determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics. A layout, which is the physical arrangement of operations created from the various processes, puts these decisions into tangible form." [p. 112]

"Customer involvement reflects the ways in which customers become part of the process and the extent of their participation." [p. 112]

"Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions." [p. 112]

"Capital intensity is the mix of equipment and human skills in a process. The greater the relative cost of equipment, the greater is the capital intensity." [p. 112]

"The process structure selected then points the way to appropriate choices on customer involvement, resource flexibility, and capital intensity." [p. 125]

"The process should reflect its desired competitive priorities. Front offices generally emphasize top quality and customization, whereas back offices are more likely to emphasize low-cost operation, consistent quality, and on-time delivery." [p. 125]

Figures 3.8, 3.9 and 3.10 (decision patterns for services and manufacturing) are reproduced in Part B Q50. (page 55)

43	What is the Six Sigma model of quality improvement? Describe the five steps of the Six Sigma improvement model. *Unit II | SEE Mar/Apr 2023 (18IM54), Q4b*	08	–	–

Answer (quoted from the textbook)

"Six Sigma A comprehensive and flexible system for achieving, sustaining, and maximizing business success by minimizing defects and variability in processes." [Glossary, p. 650]

"Black Belt An employee who reached the highest level of training in a Six Sigma program and spends all of his or her time teaching and leading teams involved in Six Sigma projects." [Glossary, p. 640]

Not in the supplied textbook: The five steps of the Six Sigma improvement model are not stated anywhere in the supplied units. The textbook index refers to the "Six Sigma Improvement Model" (book pp. 184–185, Chapter 5, "Managing Quality"), and the index entry "DMAIC process. See Six Sigma Improvement Model", but that chapter is not included, so the steps cannot be quoted.

44	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability? *Unit II | SEE Jan/Feb 2021 (18IM54), Q3a*	06	–	–

Answer (quoted from the textbook)

Same question as Part B Q31. (page 41)

45	A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis. (i) Based on these preliminary results, what percent of the time is the worker working? (ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes? *Unit II | SEE Jan/Feb 2021 (18IM54), Q3b*	10	–	–

Day	Number of times worker idle	Total number of observations
Monday	17	44
Tuesday	18	56
Wednesday	14	48
Thursday	16	60

Answer (quoted from the textbook)

"work sampling method A process that estimates the proportion of time spent by people or machines on different activities, based on observations randomized over time." [Glossary, p. 652]

(i) Percent of time the worker is working

$$\text{Idle observations} = 17 + 18 + 14 + 16 = 65$$

$$\text{Total observations} = 44 + 56 + 48 + 60 = 208$$

$$\text{Proportion idle} = \frac{65}{208} = 0.3125 = 31.25\%$$

$$\text{Proportion working} = 1 - 0.3125 = 0.6875 = 68.75\%$$

Working (textbook method applied to the numbers of the question)

Answer: The worker is working 68.75% of the time (idle 31.25%).

(ii) Additional categories for a follow-up study

Not in the supplied textbook: Not covered in the supplied textbook. The glossary definition above says only that work sampling estimates the proportion of time spent "on different activities"; the chapter text that would list follow-up categories (Chapter 4) is not in the supplied units.

46	Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings: (i) Airport (ii) Bank (iii) Classroom (iv) Product designers' office. *Unit II | SEE Jan/Feb 2021 (18IM54), Q4a*	08	–	–

Answer (quoted from the textbook)

Not in the supplied textbook: The textbook poses this as Discussion Question 6 of Chapter 3 and gives no answer; it does not list layout criteria for an airport, a bank, a classroom or a product designers' office. The statements below are the textbook's general statements on layout performance criteria.

"The goal is to improve communication among people who must interact with each other effectively, creating a good work environment." [p. 117]

"The closeness factors are indicators of the need for proximity based on an analysis of information flows and the need for face-to-face meetings." [p. 118]

"At a manufacturing plant, the closeness factor could be the number of trips (or some other measure of materials movement) between each pair of operations per day." [p. 118]

"Finally, the information gathered for OBM includes performance criteria that depend not on the relative location of department pairs, but the absolute location of a single department." [p. 118]

"When relative locations are a primary concern, such as for effective information flow, communication, material handling, and stockpicking, the weighted-distance method can be used to compare alternative block plans." [p. 118]

"Such visual representations can be particularly important when evaluating high customer-contact processes." [p. 120]

47	Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following (60 failures in total): lumps of unmixed product – 7; over or under filled jars – 18; jar lids did not seal – 6; labels rumpled or missing – 29. (No separate instruction is printed in the paper.) *Unit II | SEE Jan/Feb 2021 (18IM54), Q4b*	08	–	–

Answer (quoted from the textbook)

"process failure Any performance shortfall, such as error, delay, environmental waste, rework, and the like." [Glossary, p. 648]

"Pareto chart A bar chart on which factors are plotted along the horizontal axis in decreasing order of frequency." [Glossary, p. 647]

Pareto analysis of the 60 process failures

Process failure	Process step	Count	% of 60	Cumulative %
Labels rumpled or missing	(4) label	29	48.3	48.3
Over or under filled jars	(2) fill	18	30.0	78.3
Lumps of unmixed product	(1) mix	7	11.7	90.0
Jar lids did not seal	(3) cap	6	10.0	100.0
Total		60	100.0	

Plotted as a Pareto chart, the bars follow the order of the table (largest first): labels (29), filling (18), mixing (7), capping (6). The label step accounts for 29 of the 60 failures.

Note: The paper prints no instruction for this problem; a Pareto analysis follows from the glossary definition above. The textbook's own Pareto chart figure (Chapter 4) is not in the supplied units, so no chart is drawn.

48	Identify and discuss with block diagram the important four decisions that are essential for an effective process design. *Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3a (identical)*	08	–	–

Answer (quoted from the textbook)

The four decisions and the block diagram (Figure 3.1) are given in Part B Q40. (page 49)

49	Identify and discuss the strategies used for structuring the process in manufacturing. *Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3b (identical)*	08	–	–

Answer (quoted from the textbook)

"Process choice is the way of structuring the process by organizing resources around the process or organizing them around the products. Organizing around the process means, for example, that all milling machines are grouped together and process all products or parts needing that kind of transformation. Organizing around the product means bringing together all the different human resources and equipment needed for a specific product and dedicating them to producing just that product." [p. 115]

"The manager has four process choices, which form a continuum, to choose from: (1) job process, (2) batch process, (3) line process, and (4) continuous-flow process." [p. 115]

"A job process creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low. The workforce and equipment are flexible to handle considerable task divergence. Companies choosing job processes often bid for work. Typically, they make products to order and do not produce them ahead of time. Each new order is handled as a single unit—as a job. Examples are machining a metal casting for a customized order or producing customized cabinets." [p. 115]

"A batch process differs from the job process with respect to volume, variety, and quantity. The primary difference is that volumes are higher because the same or similar products or parts going into them are produced repeatedly." [p. 116]

"A batch process has average or moderate volumes, but process divergence is still too great to warrant dedicating a separate process for each product. The process flow is flexible, but more dominant paths emerge than at a job process, and some segments of the process have a line flow. Examples of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment." [p. 116]

"A line process lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products. Divergence is minimal in the process or line flows, and little inventory is held between the processing steps. Each step performs the same process over and over, with little variability in the products manufactured. Production and material handling equipment is specialized. Products created by a line process include the assembly of computers, automobiles, appliances, and toys." [p. 116]

"A continuous flow process is the extreme end of high-volume standardized production, with rigid line flows. Process divergence is negligible. Its name derives from the way materials move through the process. Usually, one primary material (such as a liquid, a gas, or a powder) moves without stopping through the process. A continuous-flow process differs from a line process in one important respect: Materials (be they undifferentiated or discrete) flow through the process without stopping until the whole batch is finished. The time span can be several shifts or even several months. Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food" [p. 116]

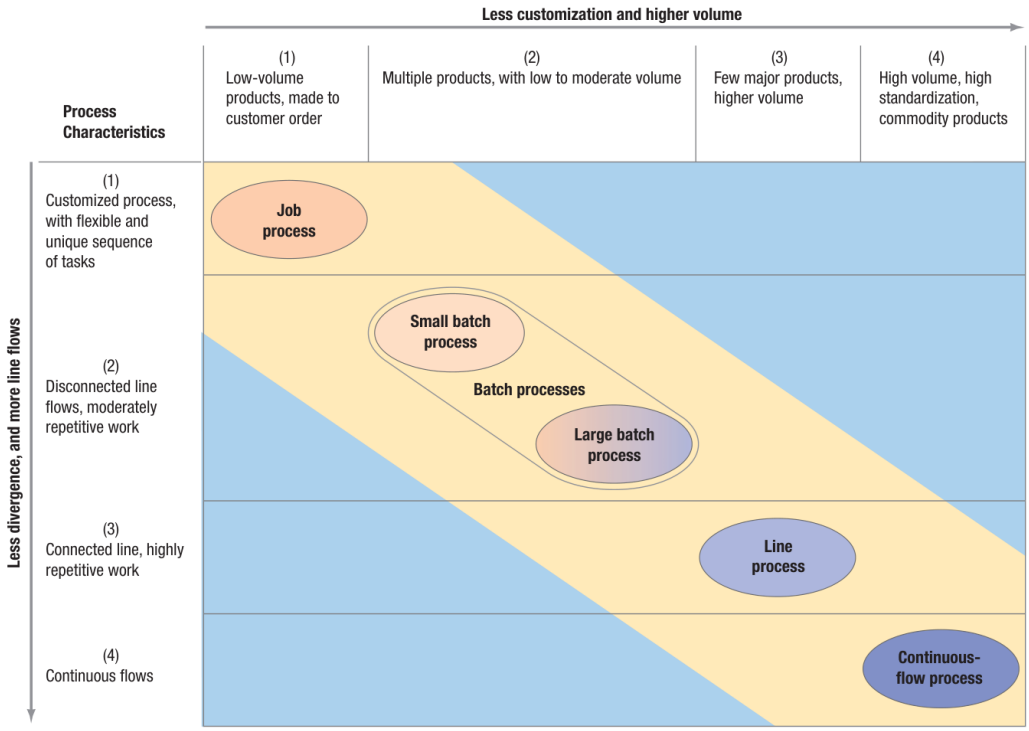


Figure 3.3: Product-Process Matrix for Manufacturing Processes (reproduced from the textbook)

The production and inventory strategies (make-to-order, assemble-to-order, make-to-stock) that go with these process choices are quoted in Part B Q32. (page 42)

50	The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit. *Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4a (identical)*	08	–	–

Answer (quoted from the textbook)

"The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. We examine services and manufacturing processes, looking for ways to test for strategic fit." [p. 124]

Services

"The process should reflect its desired competitive priorities. Front offices generally emphasize top quality and customization, whereas back offices are more likely to emphasize low-cost operation, consistent quality, and on-time delivery. The process structure selected then points the way to appropriate choices on customer involvement, resource flexibility, and capital intensity." [p. 125]

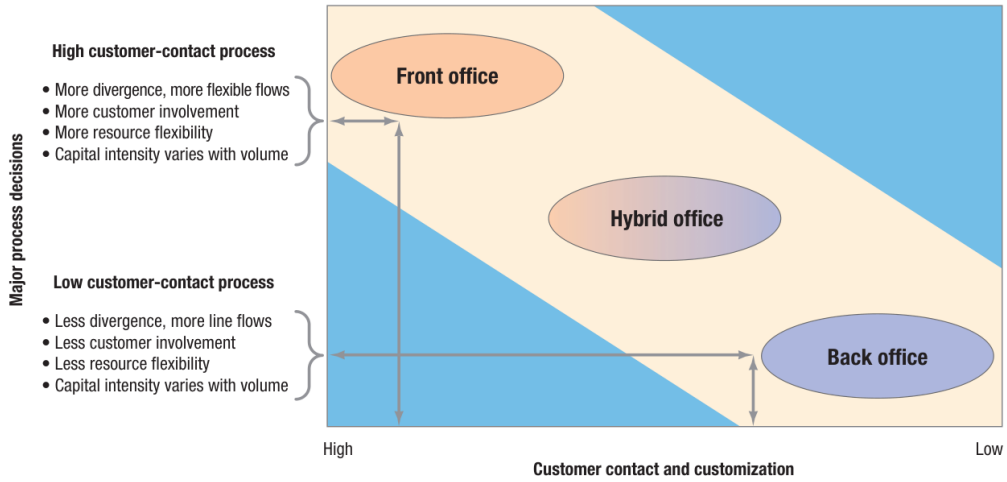


FIGURE 3.8 ▶
Decision Patterns for Service Processes

Figure 3.8: Decision Patterns for Service Processes (reproduced from the textbook)

"High customer contact at a front-office service process means: 1. Process Structure. The customer (internal or external) is present, actively involved, and receives personal attention. These conditions create processes with high divergence and flexible process flows. 2. Customer Involvement. When customer contact is high, customers are more likely to become part of the process. The service created for each customer is unique. 3. Resource Flexibility. High process divergence and flexible process flows fit with more flexibility from the process's resources—its workforce, facilities, and equipment. 4. Capital Intensity. When volume is higher, automation and capital intensity are more likely." [p. 125]

"Of course, this list provides general tendencies rather than rigid prescriptions. Exceptions can be found, but these relationships provide a way of understanding how service process decisions can be linked coherently." [p. 126]

Manufacturing

"Job and small batch processes are usual choices if top quality, on-time delivery, and flexibility (customization, variety, and volume flexibility) are given primary emphasis. Large batch, line, and continuous-flow processes match up with an emphasis on low-cost operations, consistent quality, and delivery speed." [p. 126]

"For production and inventory strategies, the make-to-order strategy matches up with flexibility (particularly customization) and top quality." [p. 126]

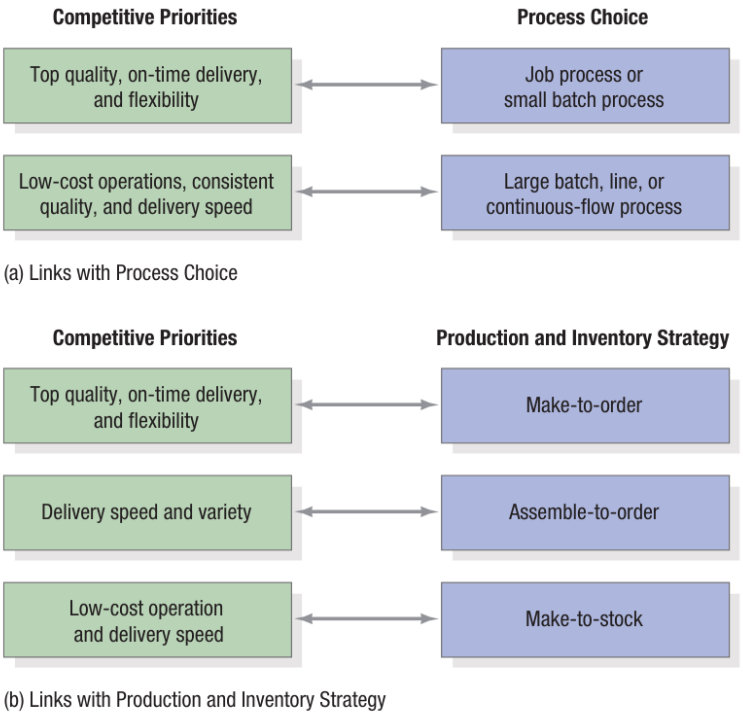


Figure 3.9: Links of Competitive Priorities with Manufacturing Strategy (reproduced from the textbook)

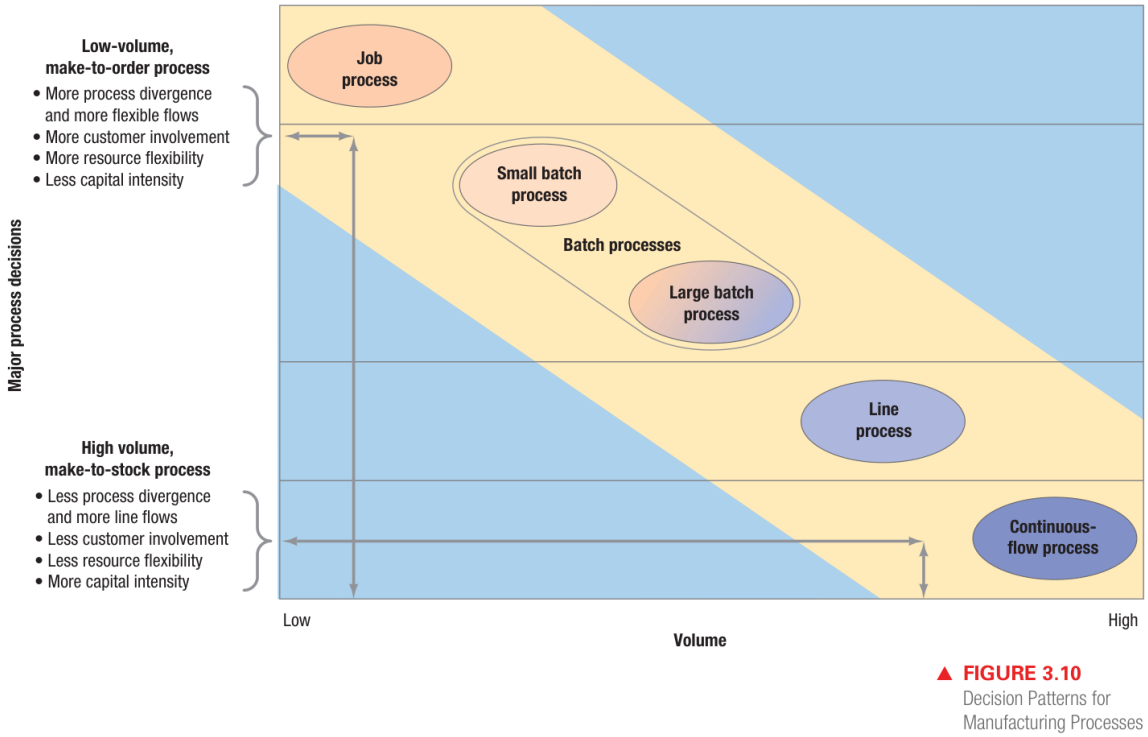


Figure 3.10: Decision Patterns for Manufacturing Processes (reproduced from the textbook)

"High volumes per part type at a manufacturing process typically mean: 1. Process Structure. High volumes, combined with a standard product, make a line flow possible. It is just the opposite where a job process produces to specific customer orders. 2. Customer Involvement. Customer involvement is not a factor in most manufacturing processes, except for choices made on product variety and customization." [p. 126]

"Resource Flexibility. When volumes are high and process divergence is low, flexibility is not needed to utilize resources effectively, and specialization can lead to more efficient processes. 4. Capital Intensity. High volumes justify the large fixed costs of an efficient operation." [p. 127]

Gaining focus

"Plants within plants (PWPs) are different operations within a facility with individualized competitive priorities, processes, and workforces under the same roof." [p. 127]

"The theory is that narrowing the range of demands on a facility will lead to better performance because management can concentrate on fewer tasks and lead a workforce toward a single goal." [p. 128]

51	Identify and discuss the key elements of process reengineering. *Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4b (identical)*	08	–	–

Answer (quoted from the textbook)

"Reengineering is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed." [p. 128]

TABLE 3.4 | KEY ELEMENTS OF REENGINEERING

Element	Description
Critical processes	The emphasis of reengineering should be on core business processes. Normal process-improvement activities can be continued with the other processes.
Strong leadership	Senior executives must provide strong leadership for reengineering to be successful. Otherwise, cynicism, resistance ("we tried that before"), and boundaries between departments can block radical changes.
Cross-functional teams	A team, consisting of members from each functional area affected by the process change, is charged with carrying out a reengineering project. Self-managing teams and employee empowerment are the rule rather than the exception.
Information technology	Information technology is a primary enabler of process engineering. Most reengineering projects design processes around information flows, such as customer order fulfillment.
Clean-slate philosophy	Reengineering requires a "clean-slate" philosophy—that is, starting with the way the customer wants to deal with the company. To ensure a customer orientation, teams begin with internal and external customer objectives for the process.
Process analysis	Despite the clean-slate philosophy, a reengineering team must understand things about the current process: what it does, how well it performs, and what factors affect it. The team must look at every procedure involved in the process throughout the organization.

Table 3.4: Key Elements of Reengineering (reproduced from the textbook)

"Reengineering has led to many successes and will continue to do so. However, it is not simple or easily done, nor is it appropriate for all processes or all organizations." [p. 128]

52	Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each: (i) Warehousing (ii) Customer service (iii) Logistics. *Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q3a*	08	–	–

Answer (quoted from the textbook)

"As we explained in Chapter 1, processes are everywhere and are the basic unit of work. Consider the following two major points: (1) supply chains have processes and (2) processes are found throughout the whole organization, and not just in operations." [p. 110]

"Part 3 is about managing supply chain processes, which are business processes that have external customers or suppliers." [p. 111]

"Processes are found in accounting, finance, human resources, management information systems, and marketing." [p. 111]

"Managers of these processes must make sure that they are adding as much customer value as possible. They must understand that many processes cut across organizational lines, regardless of whether the firm is organized along functional, product, regional, or process lines." [p. 111]

Supply chain activities (Table 3.1)

Process	Description
(i) Warehousing	Receiving shipments from suppliers, verifying quality, placing in inventory, and reporting receipt for inventory records
(ii) Customer service	Providing information to answer questions or resolve problems using automated information services as well as voice-to-voice contact with customers
(iii) Logistics	Selecting transportation mode (train, ship, truck, airplane, or pipeline) scheduling both inbound and outbound shipments, and providing intermediate inventory storage

[p. 111]

53	Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes. *Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q3b*	08	–	–

Answer (quoted from the textbook)

The block diagram of the major process decisions (Figure 3.1) and the four decisions are given in Part B Q40. (page 49)

"The key to successful process decisions is to make choices that fit the situation and that make sense together." [p. 110]

54	Define the following terms clearly: (i) Customer increment (ii) Resource flexibility (iii) Capital intensity (iv) Economies of scope. *Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q4a*	08	–	–

Answer (quoted from the textbook)

Note: The textbook term is "customer involvement"; "customer increment" in the question is taken as that term.

(i) Customer involvement

"Customer involvement reflects the ways in which customers become part of the process and the extent of their participation." [p. 112]

(ii) Resource flexibility

"Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions." [p. 112]

(iii) Capital intensity

"Capital intensity is the mix of equipment and human skills in a process. The greater the relative cost of equipment,

the greater is the capital intensity." [p. 112]

(iv) Economies of scope

"Economies of scope reflect the ability to produce multiple products more cheaply in combination than separately." [p. 124]

"If capital intensity is high, resource flexibility usually is low. In certain types of manufacturing operations, such as machining and assembly, programmable automation breaks this inverse relationship between resource flexibility and capital intensity. It makes possible both high capital intensity and high resource flexibility, creating economies of scope." [p. 124]

55	Position a process on the customer-contact matrix or product-process matrix. Bring out the characteristics of the following types of process choices: (i) Job processes (ii) Batch processes (iii) Line processes (iv) Continuous flow processes. *Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q4b*	08	–	–

Answer (quoted from the textbook)

"Figure 3.3 shows several desirable positions (often called process choices) in the product-process matrix that effectively connect the manufactured product with the process." [p. 115]

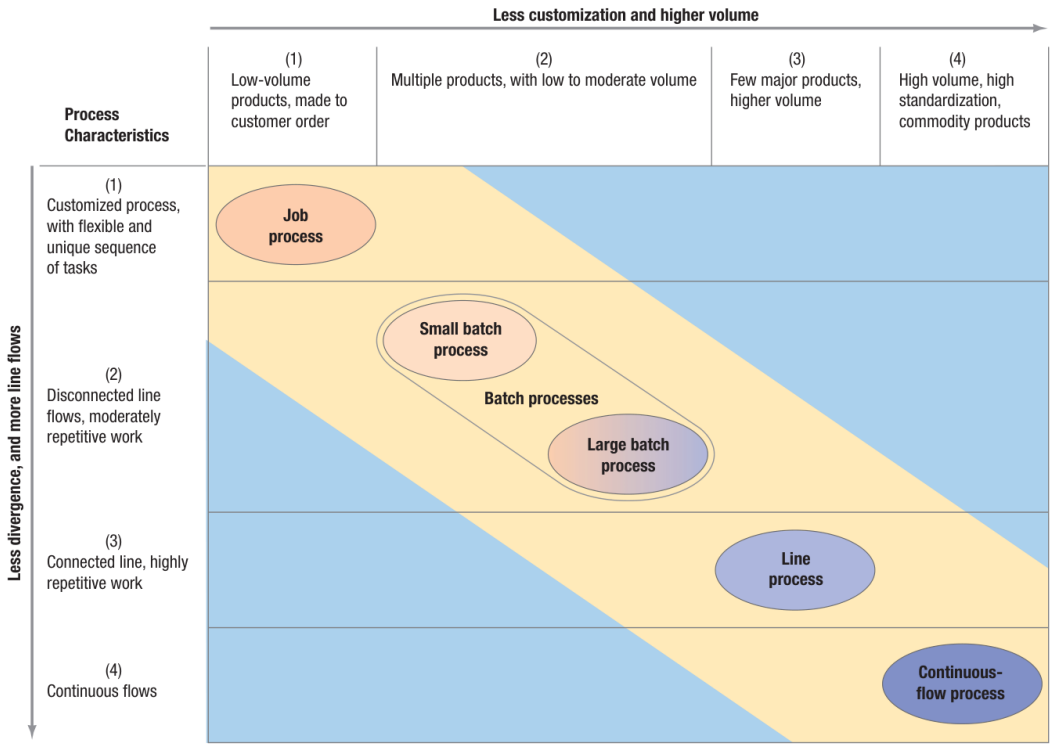


Figure 3.3: Product-Process Matrix for Manufacturing Processes (reproduced from the textbook)

"The fundamental message in Figure 3.3 is that the best choice for a manufacturing process depends on the volume and degree of customization required of the process." [p. 115]

(i) Job process

"A job process creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and volume for any one product is low. The workforce and equipment are flexible to handle considerable task divergence. Companies choosing job processes often bid for work. Typically, they make products to order and do not produce them ahead of time. Each new order

is handled as a single unit—as a job. Examples are machining a metal casting for a customized order or producing customized cabinets." [p. 115]

(ii) Batch process

"A batch process differs from the job process with respect to volume, variety, and quantity. The primary difference is that volumes are higher because the same or similar products or parts going into them are produced repeatedly." [p. 116]

"A batch process has average or moderate volumes, but process divergence is still too great to warrant dedicating a separate process for each product. The process flow is flexible, but more dominant paths emerge than at a job process, and some segments of the process have a line flow. Examples of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment." [p. 116]

(iii) Line process

"A line process lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products. Divergence is minimal in the process or line flows, and little inventory is held between the processing steps. Each step performs the same process over and over, with little variability in the products manufactured. Production and material handling equipment is specialized. Products created by a line process include the assembly of computers, automobiles, appliances, and toys." [p. 116]

(iv) Continuous-flow process

"A continuous flow process is the extreme end of high-volume standardized production, with rigid line flows. Process divergence is negligible. Its name derives from the way materials move through the process. Usually, one primary material (such as a liquid, a gas, or a powder) moves without stopping through the process. A continuous-flow process differs from a line process in one important respect: Materials (be they undifferentiated or discrete) flow through the process without stopping until the whole batch is finished. The time span can be several shifts or even several months. Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food" [p. 116]

56	Construct the product-process matrix and show in this matrix the process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly. *Unit II | SEE Fast Track Jul 2019 (16IM55/12IM62), Q3a*	08	–	–

Answer (quoted from the textbook)

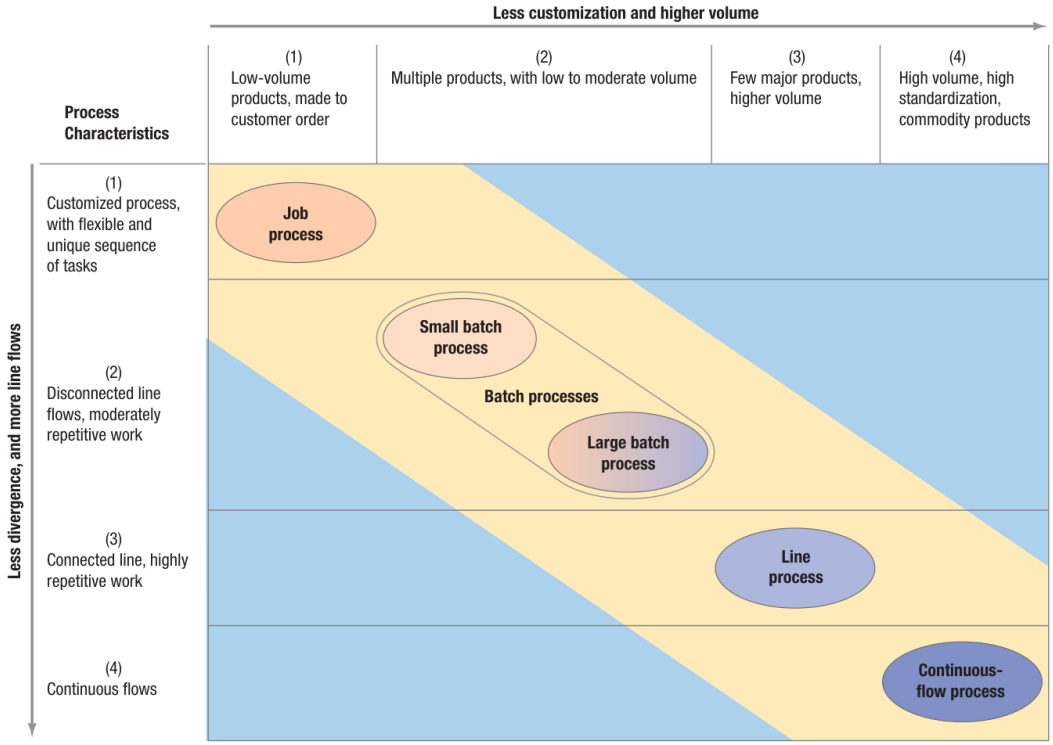


Figure 3.3: Product-Process Matrix for Manufacturing Processes (reproduced from the textbook; it shows the process characteristic categories, the level of product customization and the volume elements)

"The product-process matrix, shown in Figure 3.3, brings together three elements: (1) volume, (2) product customization, and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process itself." [p. 114]

Process structures and the organizations that use them (textbook examples)

"Examples are machining a metal casting for a customized order or producing customized cabinets." [p. 115]

"Examples of a batch process are making standard components that feed an assembly line or some processes that manufacture capital equipment." [p. 116]

"Products created by a line process include the assembly of computers, automobiles, appliances, and toys." [p. 116]

"Examples of a continuous flow process are petroleum refining; chemical processes; and processes making steel, soft drinks, and food" [p. 116]

The full description of each process choice is quoted in Part B Q55. (page 60)

57	"After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with a doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer. *Unit II | SEE Fast Track Jul 2019 (16IM55/12IM62), Q3b*	08	–	–

Answer (quoted from the textbook)

"brainstorming Letting a group of people, knowledgeable about the process, propose ideas for change by saying whatever comes to mind." [Glossary, p. 640]

"process analysis The documentation and detailed understanding of how work is performed and how it can be redesigned." [Glossary, p. 647]

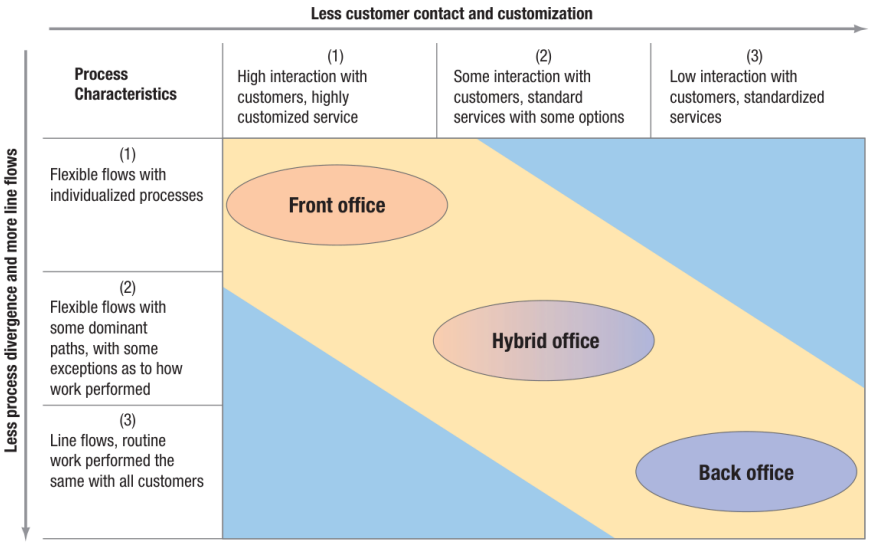
"metrics Performance measures that are established for a process and the steps within it." [Glossary, p. 646]

Not in the supplied textbook: The passage quoted in the question, the doctor-and-patient comparison, the role of questioning, and the worked example are in Chapter 4 (book pp. 139–160, index entries "Brainstorming, 155–157"), which is not in the supplied units. Only the glossary definitions above are available, so no further textbook answer can be quoted.

58	Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate on the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process? *Unit II | SEE Fast Track Jul 2019 (16IM55/12IM62), Q4a*	08	–	–

Answer (quoted from the textbook)

"The customer-contact matrix, shown in Figure 3.2, brings together three elements: (1) the degree of customer contact, (2) customization, and (3) process characteristics. The matrix is the starting point for evaluating and improving a process." [p. 113]



◀ **FIGURE 3.2**
Customer-Contact Matrix for Service Processes

Figure 3.2: Customer-Contact Matrix for Service Processes (reproduced from the textbook)

Degree of customer contact and customization

"The horizontal dimension of the matrix represents the service provided to the customer in terms of customer contact and competitive priorities. A key competitive priority is how much customization is needed. Positions on the left side of the matrix represent high customer contact and highly customized services." [p. 113]

Process divergence and flow

"The vertical dimension of the customer-contact matrix deals with two characteristics of the process itself: (1) process divergence and (2) flow. Each process can be analyzed on these two dimensions." [p. 113]

"Process divergence is the extent to which the process is highly customized with considerable latitude as to how its tasks are performed. If the process changes with each customer, virtually every performance of the service is unique. Examples of highly divergent service processes where many steps in them change with each customer are found in consulting, law, and architecture." [p. 113]

"A service with low divergence, on the other hand, is repetitive and standardized. The work is performed exactly the same with all customers, and tends to be less complex." [p. 113]

"A flexible flow means that the customers, materials, or information move in diverse ways, with the path of one customer or job often crisscrossing the path that the next one takes." [p. 113]

"A line flow means that the customers, materials, or information move linearly from one operation to the next, according to a fixed sequence." [p. 114]

Starting point for evaluating and improving a process

"After analyzing a process and determining its position on the customer-contact matrix in Figure 3.2, it may be apparent that it is improperly positioned, either too far to the left or right, or too far to the top or bottom. Opportunities for improvement become apparent. Perhaps, more customization and customer contact is needed than the process currently provides. Perhaps, instead, the process is too divergent, with unnecessarily flexible flows. Reducing divergence might reduce costs and improve productivity." [p. 125]

59	Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating a manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for fixed and flexible automation respectively. Justify automation in the context of capital intensity. *Unit II | SEE Fast Track Jul 2019 (16IM55/12IM62), Q4b*	08	–	–

Answer (quoted from the textbook)

"Capital intensity is the mix of equipment and human skills in the process; the greater the relative cost of equipment, the greater is the capital intensity." [p. 122]

"Automation is a system, process, or piece of equipment that is self-acting and self-regulating. Although automation is often thought to be necessary to gain competitive advantage, it has both advantages and disadvantages. Thus, the automation decision requires careful examination." [p. 123]

Advantages

"Substituting labor-saving capital equipment and technology for labor has been a classic way of improving productivity and quality consistency in manufacturing processes." [p. 123]

"However, fixed automation maximizes efficiency and yields the lowest variable cost per unit if volumes are high." [p. 123]

Disadvantages

"One big disadvantage of capital intensity can be the prohibitive investment cost for low-volume operations (see Figure 3.7). Generally, capital-intensive operations must have high utilization to be justifiable. Also, automation does not always align with a company's competitive priorities. If a firm offers a unique product or high-quality service, competitive priorities may indicate the need for hand labor and individual attention rather than new technology." [p. 123]

Fixed automation

"Manufacturers use two types of automation: (1) fixed and (2) flexible (or programmable). Particularly appropriate for line and continuous-flow process choices, fixed automation produces one type of part or product in a fixed sequence of simple operations. Operations managers favor fixed automation when demand volumes are high, product designs

are stable, and product life cycles are long. These conditions compensate for the process's two primary drawbacks: (1) large initial investment cost and (2) relative inflexibility." [p. 123]

Flexible automation

"Flexible (or programmable) automation can be changed easily to handle various products. The ability to reprogram machines is useful for both low-customization and high-customization processes." [p. 123]

"An industrial robot, which is a versatile, computer-controlled machine programmed to perform various tasks, is a classic example of flexible automation." [p. 123]

Examples in the textbook

"Gillette, for example, spent $750 million on the production lines and robotics that gave it a capacity to make 1.2 billion razor cartridges a year. The equipment is complicated and expensive. Only with such high volumes could this line process produce the product at a price low enough that consumers could afford to buy it." [p. 123]

"It uses a line process dedicated strictly to Phaeton cars. Because of its focus on Phaetons, which are made nowhere else, it enjoys high volumes. This volume justifies the high automation invested in this $208 million plant." [p. 123]

"Flexible Automation at Just Born, Inc., a candy company in Pennsylvania" [p. 124]

"The robot is regularly reprogrammed based on the marshmallow configuration being produced." [p. 124]

Note: The textbook does not name its examples of high-volume automated line processes (Gillette, Volkswagen Phaeton) as "fixed automation"; it labels the Just Born robots and the industrial robot as flexible automation.

Automation in the context of capital intensity

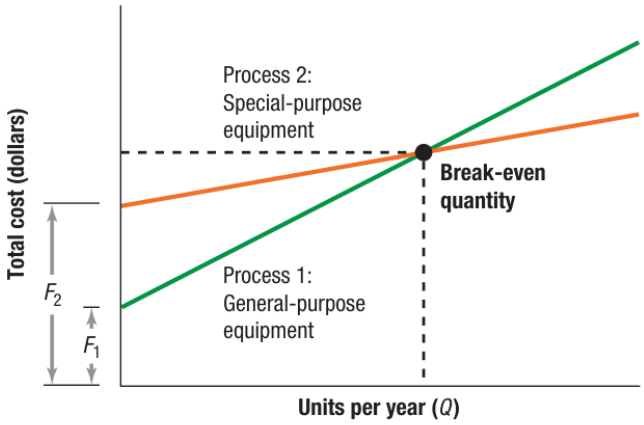


Figure 3.7: Relationship Between Process Costs and Product Volume (reproduced from the textbook)

"Low volumes mean that process designers should select flexible, general-purpose equipment." [p. 122]

"Conversely, process 2 is the better choice when volumes are high and customization is low. Its advantage is low variable unit cost, as reflected in the flatter total cost line. This efficiency is possible when customization is low because the equipment can be designed for a narrow range of products or tasks. Its disadvantage is high equipment investment and, thus, high fixed costs (F2). When annual volume produced is high enough, spreading these fixed costs over more units produced, the advantage of low variable costs more than compensates for the high fixed costs." [p. 122]

"If capital intensity is high, resource flexibility usually is low." [p. 124]